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Exploring the Integration of Pumped Hydro Energy Storage on the Operation of the Eastern Røldal-Suldal Hydropower System

Karoline Josefine Gudbjerg
Environmental Physics and Renewable Energy

Abstract

Significant investments in renewable energy sources are being made globally [1]. Integrating wind and solar power, which are intermittent energy sources, presents a substantial challenge for power systems. The mismatch between energy generation and demand fluctuates [2]. Addressing this challenge requires developing flexibility within the system to effectively manage these unregulated energy sources [3]. Pumped Hydro Energy Storage (PHES) is a regulative renewable energy source which offers long- and short-term energy storage [2]. As an environmentally sustainable resource, hydropower enhances flexibility and storage capacity within the energy grid. These attributes are crucial for improving grid stability, supporting the integration of intermittent renewable sources and facilitating load balancing [4].

This thesis explores the implementation of PHES in the eastern part of the Røldal-Suldal power system (RSK). Furthermore, it examines the effects of different pump sizes on the operational dynamics of the system. The program used for simulating the hydropower system is ProdRisk, which employs a combination of Stochastic Dual Dynamic Programming (SDDP) and Stochastic Dynamic Programming (SDP). These methods are ideal for optimizing hydropower scheduling [5]. The simulations of the RSK system were run on the ProdRisk portal using the code language python. The simulations examined pump sizes of 10 MW, 20 MW, 30 MW, and 40 MW over a 30-year period, utilizing simulated electricity prices from the EMPS model.

In summary, implementing a pump in the eastern part of RSK is projected to increase income from electricity sales compared to a system without a pump. A plant equipped with a bigger pump has a longer operation time, both for the turbine and pump. This is particularly notable during years of low inflow. While a turbine with a larger pump may have lower efficiency, the pump empowers the turbine to generate more electricity when the demand is high. The annual mean production is increasing from 962.9 GWh in the baseline simulation (0 MW) to 1000 GWh for the 40 MW pump simulation. However, annual net production declines with larger pumps, dropping from 962.9 GWh in the baseline simulation to 955 GWh for the 40 MW pump simulation. Moreover, revenue from electricity sales increases by 2.56% for the largest pump of 40 MW compared to a baseline simulation.

Overall, this thesis demonstrates the application of ProdRisk and the beneficial impacts of PHES implementation. Utilizing PHES could be crucial in enhancing the stability and flexibility of future power systems, while also helping to ensure that global temperatures remain below 1.5°C.

Sammendrag

Globalt blir det gjort store investeringer i fornybare energikilder [1]. Integrering av periodiske energikilder som vind og solkraft representerer en stor utfordring for kraftsystmene. De periodriske energikildene fører til svigninger i balansen mellom enrgiproduksjon og forbruk [2]. For å møte denne utfordringen er det nødvendig å utvikle fleksibilitet i systemet for effektiv håndtering av de uregulerte energikildene [3]. Pumpetkraft er en regulert fornybar energikilde som tilbyr både kort- og langtidslagring av energi [2]. Pumpekraft er en fornybar resurss som bidrar til å øke fleksibiliteten og lagringskapasiteten i energinettet. Disse egenskapene er avgjørende for å styrke nettstabiliteten, muliggjøre integrasjon av periodiske fornybare energikilder, og bidra til balansering av lasten i nettet [4].
Denne masteroppgaven utforsker implementeringen av pumpekraft i den østlige delen av Røldal-Suldal kraftsystem (RSK). Videre undersøkes det hvordan ulike pumpestørrelser påvirker driftsdynamikken i systemet. Vannkraftsystemet blir simulert i programmet ProdRisk. ProdRisk baserer seg på en kombinasjon av Stokastisk Dual Dynamisk Programmering (SDDP) og Stokastisk Dynamisk Programmering (SDP). Disse metodene er ideelle for optimalisering av vannkraftplanlegging [5]. Simuleringene av RSK-systemet ble utført i ProdRisk-portalen ved bruk av programmeringsspråket Python. Pumpestørrelsene var 10 MW, 20 MW, 30 MW og 40 MW og ble simulert over en periode på 30 år. Strømprisene var simulert av markedsmodelen EMPS.

Oppsummert viser simuleringene at implementering av en pumpe i østlige del av RSK forventes å øke inntektene fra elektrisitetssalg sammenlignet med systemet uten pumpe. En kraftstasjon utstyrt med større pump oppnår lengre driftsperioder for både turbinen og pumpen. Driftsperioden for både turbinen og pumpen øker i år med lav tilsig. En større pumpe har lavere virkningsgrad men den gir kraftverket mulighet til å produsere mer strøm når etterspørselen er høy. Den årlig gjennomsnittsprodusjonen Den årlige gjennomsnittsproduksjonen øker fra 962,9 GWh i basissimuleringen uten pumpe til 1000 GWh for 40 MW pumpe-simuleringen. Imidlertid avtar den årlige nettoproduksjonen med større pumper, og faller fra 962,9 GWh i basissimuleringen til 955 GWh for 40 MW pumpe-simuleringen. I tillegg øker inntektene fra strømsalg med 2,5% for den største pumpen på 40 MW sammenlignet med en basissimulering uten pumpe.

Masteroppgaven viser gjennom anvendelse av ProdRisk de gunstige virkningen av å implementere PHES. Bruk av pumpekraft kan være avgjørende for å styrke stabiliteten og fleksibiliteten i fremtidige kraftsystemer, samtidig som det bidrar til å sikre at globale temperaturer forblir under 1,5°C.

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1 Introduction

1.1 A transforming energy market with the need for storing energy

Worldwide, power grids are swiftly shifting from fossil fuel-based systems to embrace renewable energy sources [1]. The European energy system and the Norwegian sector are progressing towards achieving carbon neutrality. According to the Norwegian Transmission System Operator Statnett's long-term market analysis for 2022–2050, the electrification of society results in a significant increase in demand for power. The report states that future energy will mainly come from wind and solar power [3]. Significant investments in solar and wind power are being made in several countries worldwide [1]. These sources are characterized by their variability and lack of regulation in the power output, leading to a mismatch between energy generation and variations in energy demand [2]. To smooth out the considerable fluctuations in generation, it is essential to foster the development of substantial flexibility [3]. Flexibility refers to the capability of producers, consumers and energy storage systems to modify their energy production, consumption and storage in response to changes in the power system. Flexible consumption can be rescheduled, reduced or completely shut down for varying durations [6]. One form of flexibility is large-scale energy storage.

The year 2023 experienced the most substantial rise in the capacity of renewable energy sources. The world collectively added nearly 473 GW of renewable energy capacity. That is an increase in renewable energy by 13.9% from the previous year, 2022 [7]. Also, 86% of the newly added power generation capacity worldwide came from renewable energy sources. The growth in renewable energy is mainly due to the increased expansion of solar and wind power. Solar power made up three-quarters of the additional renewable capacity, with 346 GW of newly installed capacity, whereas wind energy had an addition of 116 GW. The annual renewable capacity additions need to be 1050 GW for the rest of this decade to meet the target of limiting the rise in global temperature with 1.5°C [7].

The transition in the power system towards renewable energy sources introduces new challenges in maintaining supply security and managing power, frequency, and voltage regulation. Pumped Hydro Energy Storage (PHES) is a technology that can deliver short-term and long-term energy storage [2]. PHES involves the process of pumping water from a lower reservoir to an upper reservoir during periods of surplus power generation, such as windy and sunny days. Subsequently, the stored water is released from the upper reservoir, flowing back to the lower reservoir through a turbine to generate electricity during times of insufficient supply. At times, utilizing PHES can be advantageous, especially when there is surplus of power capacity and

the electricity prices are low [1]. PHES is the world's largest energy storage technology both in energy volume and in power capacity [2, 1].

Numerous PHES systems have been strategically developed in tandem with conventional hydroelectric systems situated along rivers. In countries with hydropower, it plays a significant role. By varying the flow of water through a turbine, hydropower effortlessly adapts to variable loads in the power system [1]. In addition to being environmentally sustainable, hydropower offers flexibility and storage capacity. These qualities make it possible to improve grid stability and facilitate the implementation of periodic renewable energy sources such as wind and solar and offer load balancing in the grid [4]. Additionally, PHES has fast response speed and flexible start and stop [8]. PHES provides several ancillary services to the power grid, including voltage stability preservation and frequency variation, among others [4]. More details on the characteristics and services of PHES can be found in the theory section 3.1. Lastly, as mentioned in the previous paragraph, PHES is environmentally sustainable. Unlike wind and solar power, PHES is a renewable energy source that can be regulated. Additionally, because PHES has developed in conjunction with conventional hydropower systems, expanding PHES has a smaller environmental impact compared to building wind and solar farms. Moreover, a key advantage of PHES is its ability to integrate fluctuating renewable energy sources such as wind and solar into the grid. This capability positions PHES as a prime candidate for investment in renewables, due to its flexibility and storage properties that enhance grid stability. Additionally, PHES emerges as a key solution for achieving the goal of limiting global warming to 1.5°C [4, 1].

1.2 Norway and its hydropower history

Reservoirs for hydropower in Norway comprise approximately 50% of Europe's total energy storage capacity within hydropower systems. The country has 87 TWh of energy storage in reservoirs, with 10 to 20 TWh accessible most of the time. The Norwegian hydropower reservoirs have a large potential to enhance energy storage capabilities on a European scale by expanding their pumped storage capacity. Norway is the sixth largest in the world in generating renewable energy from hydropower [2]. Norway generates a total of 156.9 TWh of energy annually, with a significant 136.9 TWh coming from hydropower. In 2023, hydropower accounted for 87.3% of Norway's energy production, utilizing its 1781 power stations [9].

There are ten existing pumped storage plants in Norway with a total capacity of 1369 MW and an energy storage capacity of 5 TWh. Despite the total installed hydropower capacity being 33 852 MW, PHES remains underexploited in Norway, especially considering the prominence of hydropower in the country. The PHES are

located in the west and central parts of Norway. The PHES facilities were commissioned between 1955 and 2005 and all are open-loop systems with natural inflow. These facilities have a net energy production of 1.8 TWh per year, where gross energy is 2.6 TWh and 0.8 TWh is used for pumping. In Norway, the design of pumped storage plants is specifically tailored for seasonal use. Consequently, the capacity of the upper reservoirs is significantly larger than that of the lower reservoirs across these facilities [2].

The development of hydropower and PHES was strongly influenced by the industry development in Norway. In 1991, the Energy Act was adopted in Norway. The law changed the organization of power supply and remains valid today. Before the law, local authorities and parliament (Stortinget) decided the electricity prices. The law facilitated the implementation of market-based principles for generating and selling electrical power. The electrical grid, due to its unique status as a monopoly, was placed under specific regulatory oversight. The production and sale of electrical power were subjected to competition. The Energy Act provided the legal framework necessary for optimizing the power market. Also, The Energy Act ensuring that production is maximized and developed in a manner that best serves socio-economic interests. After the introduction of the Energy Act, the development of hydropower stations and PHES stations significantly decreased. The main reason was supply exceeded power demand. Due to the combined factors of power producers being required to ensure power supply in their designated areas and the limited market flexibility, there was a drive towards investing in excess capacity. This is also one of the reasons why Norwegian pumped storage is designed for seasonal pumping and not for short-term pumping. Furthermore, due to Norway's abundant hydropower resources and large energy reserves, there is less need to invest in expensive system services to maintain energy balance and grid stability. There has not been a need in the Norwegian power system for PHES because of the access to power and energy reserves due to the huge number of reservoirs in Norway [10].

1.3 The Nordic power system and power prices

The Nordic power system is closely interconnected. Norway has seven connections to Sweden, one to Finland, four to Denmark, and one to The Netherlands called NorNed. Recently, Norway built a connection to Germany called NordLink and a connection to Great Britain called North Sea Link. The connections were established in March 2021 and September 2022, respectively [11, 12]. The power exchange operates such that power flows from areas with low prices to areas with high prices. The advantages of the power trade are income to Norwegian power producers and increased power supply for Norwegian consumers. If the power prices are lower in the other trading countries, Norway will import electricity and conserve water in

reservoirs for future use [13].

The Norwegian and the Nordic power system is strongly influenced by the European energy transformation through electricity prices, the EU Emissions Trading System, power exchange and new regulations. Additionally, increase in demand and cheaper technology influence the electricity prices.

The Nordic power consumption will see a substantial increase by 2050, driven by electrification and increased industrial activity. Additionally, power production is expected to double. Overall, in the Nordic countries, wind power will be the leading supplier of power, with contributions from solar, hydro and nuclear power. These trends are anticipated to continue from the current situation through 2050. According to Statnett [3], the Nordic power consumption will increase to 850 TWh in 2050, this amount represents a doubling from the 400 TWh recorded in 2023. Further on, in Norway, consumption is increasing significantly. This rise is driven by efforts to reduce emissions through electrification and the development of new industries and commercial activities. The Norwegian consumption will increase to 220 GWh in 2050, a rise of 80 TWh from today's consumption. Solar and hydropower will have to increase their generation by 30 TWh by 2050. Moreover, offshore wind will deliver 40 TWh by 2050 [3].

The power market in Europe will have high short-term volatility in the first five to ten years. By 2035–2040, the expansion of more affordable flexibility will lower price volatility [3]. According to Statnett, there are several factors for the high electricity prices. The first reason is the high prices of gas and coal due to the ongoing war in Ukraine. The war influences both the power market and the supply in the rest of the world. The weather is also an important factor. As the reservoir level decrease, the value of the remaining water increase , which correspondingly affects the spot price. There has also been much debate on how much the two new cross-border connections have impacted the power prices in Norway [13]. The North Sea Link is a 720 km deep-sea cable with a capacity of 1400 MW between the Suldal municipality in Norway and the Newcastle area in England. NordLink is also a deep-sea cable between Sirdal municipality in Norway and Wilster in Germany. The cable is 623 km and has a capacity of 1400 MW [11, 12]. According to studies by Statnett, the price impact of the two new power cables to Germany and England accounts for about 10% of the unusually high electricity prices since the autumn of 2021. Even without these cables, electricity prices in Southern Norway would have been at record highs [14]. Additionally, the prices of CO_2 quotas have risen significantly from 2017 to the present. In 2017, the cost of one ton of CO_2 was approximately 5 euros, peaking at around 97 euros per ton in 2023. As of 2024, the price stands at about 70 euros per ton [15]. This increase has led to high power prices in Europe and in the Nordic countries. CO_2 quotas are an important instrument to make the expansion of wind

and solar power more profitable, and they enhance the competitive edge of renewable energy sources [13].

Norway is divided into five electricity price zones: Østlandet (NO1), Sørlandet (NO2), Midt-Norge (NO3), Nord-Norge (NO4) and Vestlandet (NO5). The Norwegian power market, driven by the weather and natural energy sources, varies significantly across these regions. Price zones are primarily determined by bottlenecks in the transmission network, where the capacity to transfer power is limited. For instantnce, the transportation of electricity between Nord-Norge (NO4) and Østlandet (NO1) passes through Sweden. However, between Vestlandet (NO5) and Østlandet (NO1), the transmission capacity is substantial due to the high demand for power transfer. This stems from extensive power production in western Norway and comparatively lower production in the east. Moreover, the east, around Oslo, is home to the majority of Norway's population. The configuration of these price zones not only indicates the balance of power scarcity or surplus but also ensures the power system's safe operation [16].

1.3.1 Hydropower scheduling

Due to significant price volatility and the increasing demand for large-scale energy storage, there is a substantial need for the development of PHES systems. These systems are capable of providing both short-term and long-term solutions. Given the interconnectivity with the rest of Europe, questions are arising about the necessity and profitability of PHES in Norway.

Hydropower is widely used in Norway. In 2023, there were 1781 hydropower stations scattered throughout the country, with a significant concentration in the western part. The Norges vassdrags- og energidirektorat, NVE, maps of power stations [17], in Norway illustrate a complex network, where many plants utilize the same river systems and are interconnected within the same system. Furthermore, a hydropower system may comprise one or several plants, along with reservoirs and a complex network of rivers. Numerous factors must be considered to achieve the most efficient and sustainable operation of hydropower system [18]. Additionally, hydropower stations in Norway sell power to various electricity markets, including the spot and reserve markets. These markets are expected to operate at shorter intervals. This means that periods during prices are established, trades are executed, data is collected and distributed will be shorter than current intervals. This will require hydropower operators to respond more quickly to changes in demand. Consequently, producers require advanced tools to support their decision-making process, ensuring that they maximize the value extracted from the water resources stored in their reservoirs [18].

Scheduling hydropower generation is a specialized area of research. It involves solving

complex mathematical problems and requires technical knowledge of operating crucial components such as turbines, generators, and water flow. Moreover, it includes forecasting inflows and market prices, applicable to both long-term and short-term time resolution [19].

The research organization SINTEF has developed several models to optimize hydropower scheduling. The first model, EMPS, focuses on power system analysis and long-term scheduling. The second, which is employed in this thesis, is ProdRisk, a model for long-term hydropower scheduling optimization. The third is the SHOP model, designed for short-term daily market engagement to ensure optimal operation. The majority of Nordic hydropower producers utilize this range of models to ensure the efficient management of their systems [18].

1.3.2 Day-ahead electricity prices and scenarios of future prices

Electricity price zones, also known as price areas, form the market framework for trading electricity. Electricity prices are set for each specific area on an hourly basis for the next 24 hours. These prices are not government-regulated but result from the supply and demand for power within each price area, as reported to NordPool, the Nordic power exchange. NordPool is owned by the Nordic system operators [20].

The day-ahead market at NordPool functions as an auction-based exchange where electricity is physically delivered. In this market, producers and consumers submit bids and offers, allowing NordPool to calculate an hourly price that balances these factors. NordPool then sets the price for each hour of the next day. Every morning, customers place their bids in the auction for the following day, specifying the amount of electricity in megawatts they are willing to buy or sell at predetermined price levels (EUR/MWh) for each hour of the day [21].

Moreover, intraday markets facilitate the trading of electricity closer to the time of delivery. These markets operate simultaneously with the day-ahead market. This arrangement helps maintain a balanced supply and demand. As the production of renewable energy, which can be intermittent, increases, there is a growing interest in participating in the intraday markets. This interest originates from the challenges that market participants face in maintaining balance after the day-ahead market closes. The intraday market operates continuously, 24/7, allowing trading up until one hour before delivery, and in some cases, even closer to the delivery hour. Prices are determined on a first-come, first-served basis, with transactions prioritizing the most favorable prices, the highest buying prices and the lowest selling prices [22].

As power systems increasingly rely on intermittent renewable energy sources, energy supply becomes more unpredictable, leading to greater fluctuations in electricity

prices [3]. Furthermore, as previously mentioned, renewable sources pose challenges in power planning due to their non-regulative nature. To enhance predictions of power demand, it is crucial to employ market models that can provide forecasts about future market conditions.

To accurately forecast future electricity prices, one can utilize the EMPS model. The EMPS model, EFI's Multi-area Power-market Simulator model, is a market model [23]. The model is a software for optimizing and simulating hydrothermal power systems. It is widely used in deregulated regions to forecast future spot prices. Additionally, the model is adept at providing forecasts for various aspects such as production, reservoir levels, water flow, consumption, rationing, import and export. In the model each price scenario is linked to a specific weather year. The EMPS model is used for forecasting future electricity prices [24].

1.3.3 Electricity Reserve Markets

As previously mentioned, one of the benefits of PHES, given its characteristics, is its ability to regulate power-grid frequency. Frequency stability in the power system is reliant on the rotating mass of synchronous machines connected to the grid. Most of this rotating mass is located in hydro and thermal power plants.

The power system must maintain equilibrium at all times, ensuring that the sum of production and import consistently matches the sum of consumption, export and losses. Maintaining this balance is crucial to keep the system's frequency around 50 Hz. This is a standard requirement of the Nordic power system. Reserves and flexibility services, either through additional power production or consumption, are activated when the frequency deviates from this norm. If the frequency rises above 50 Hz, it indicates that supply exceeds demand, conversely, if it falls below 50 Hz, demand outstrips supply. Statnett, Norway's transmission system operator is responsible for maintaining this balance. The most critical tool for managing these imbalances is purchasing flexibility from the electricity reserve market. In Norway, there are four types of reserves: primary, secondary, tertiary and fast frequency reserves. Each type has distinct aims and characteristics, but all are designed to help maintain system balance.

The first line of response to frequency changes in the system is the fast frequency reserves, also called Fast Frequency Reserves (FFR). These changes occur when the systems' inertia, which is the ability to resist changes, falls below a certain level. The primary goal of FFR is to halt these frequency shifts. Following the initial response, the primary reserves, known as Frequency Containment Reserves (FCR), take over. The purpose of FCR is to stabilize the frequency at a new level. After stabilization, the secondary reserves, or Automatic Frequency Restoration Reserves (aFRR), kick

in. Their objective is to return the frequency to its normal range. Finally, the tertiary reserves, the Manual Frequency Restoration Reserves (mFRR), maintain the balance until a new equilibrium is achieved in the power market [25].

Due to the widespread utilization of hydropower and significant reservoir capacity in Norway, there is growing interest in exploring the necessity and profitability of PHES systems in the country. Interest is particularly strong in integrating PHES with existing hydropower facilities [26]. The Røldal-Suldal hydropower system is a prominent example of such integration.

Given the extensive use of hydropower in Norway, coupled with enormous reservoir capacities, there are increasing intrests about the necessity and profitability of PHES systems in the country. Consequently, several conventional hydropower systems are being evaluated for potential integration with PHES technologies. One such system under consideration is the Røldal-Suldal hydropower system.

1.4 Case Study: Eastern Røldal-Suldal hydropower system

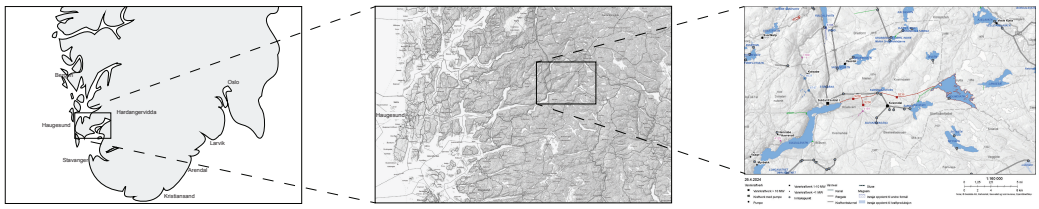


Figure 1: Geographical location of the Røldal-Suldal Power stations (RSK) in Norway. The figure have been adapted and modified from [17]

The area being investigated in the thesis is the eastern part of the hydropower system Røldal-Suldal (RSK). The geographical location of RSK is shown in figure 1. RSK extends along Røldal and Suldal watercourse and ends at Suldalsvannet [27]. The catchment area for the power plants in Røldal-Suldal is 790 km^2 [28]. The hydropower system consists of nine power stations: Suldal I, Suldal II, Røldal, Novle, Kvanndal, Svandalsflona, Middyr, Vasstøl and Midtlæger. The Røldal-Suldal hydropower complex is owned by Lyse Kraft DA, a Norwegian hydropower company. Lyse Kraft Da is Norway's third-biggest hydropower producer and is 25.6% owned by Hydro and 74.4% by Lyse. Hydro is the operator of the complex, and Lyse is responsible for both market operations and management of water resources [27].

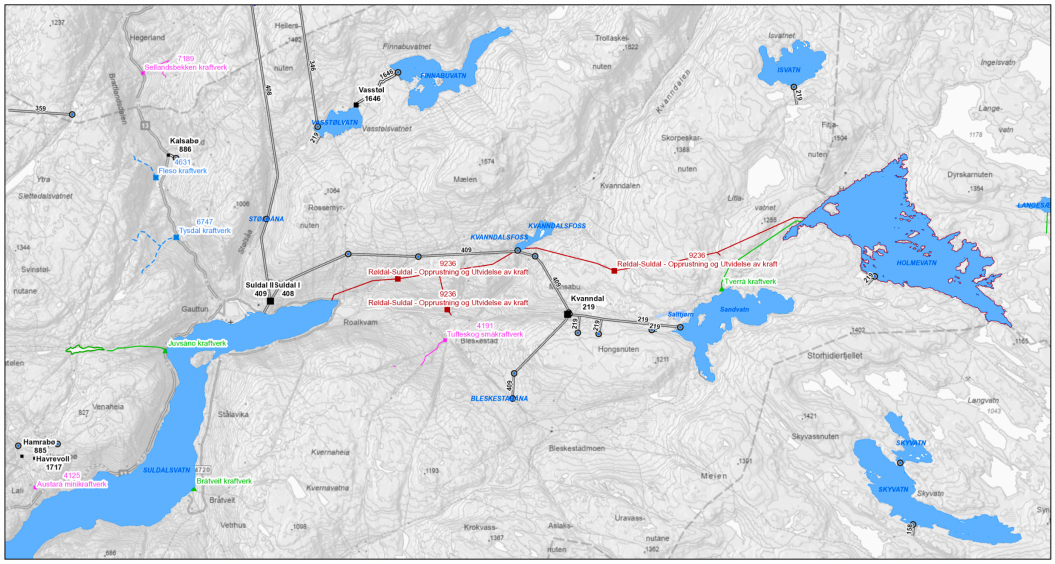


Figure 2: Map of the eastern part of RSK. Isvann, the northernmost reservoir on the map, is situated above Holmevann which is the largest reservoir and lies to its south. The water flows from Isvann and Holmevann down to Sandvann, continuing on to the Kvanndal power station. Sandvann and Kvanndal are located southwest of Holmevann. The water then proceeds from Kvanndal to the Kvanndalsfoss, through Suldal II power station before it finally reaches Suldalsvannet. Kvanndalsfoss is positioned northwest of Kvanndal, while the Suldal II power station is southwest of Kvanndalsfoss [17].

The eastern part of RSK consists of four regulative reservoirs and two hydropower stations, shown in figure 2. In the Eastern watercourse, water is harnessed from Holmevann, varying between 1058 and 1048 meters above sea level (masl), from Isvann, ranging from 1295 to 1985 masl, and it descends to Suldalsvann, situated at 68 masl. Along this route, the water is first collected in Sandvann reservoir, with levels ranging from 950 to 924 masl, before passing through the Kvanndal and Suldal II power stations. Additionally, Suldal II receives water from the Kvanndalsfoss reservoir, which has a level variance between 630 and 620 masl [17].

The Kvanndal and Suldal II power stations are located in the Suldal municipality in Rogaland County. Kvanndal, established in 1967 as the first station in the Eastern scheme, features a Francis turbine in its single generator, capable of producing up to 40 MW. On average, it generates 188.9 GWh per year. It has a gross head of 314 meters and can handle a flow of up to 15 m^3/s [29, 30]. Suldal II, also started in 1967, has a higher capacity, with a maximum output of 150 MW and an annual average production of 738.6 GWh. This station includes two generators, each with its own Francis turbine. It operates with a gross head of 559 meters and can manage a maximum flow of 31.5 m^3/s [31, 30].

Moreover, the four reservoirs in the Eastern watercourse vary significantly in size. Holmevann is the largest, boasting a maximum capacity of 96 Mm^3, then Sandvann with 66 Mm^3 then Isvann at 16 Mm^3, and finally, Kvandalsfoss, which holds up to 1.6 Mm^3 of full volume [17].

Lastly, in March 2024, Lyse DA submitted a concession application for the Kvanndal II PHES. This indicates a significant interest in investing in PHES within the RSK system. [32].

1.5 Research question

This master's thesis focuses on the Røldal-Suldal power (RSK) stations located in the southwest part of Norway, exploring the implementation of PHES in this system. The central research question is:

How does employing PHES with the SDDP method impact the eastern Røldal-Suldal hydropower system's operations? How do pump size variations affect the Kvanndal power plant's efficiency and performance?

The thesis aims to identify the operational impacts of PHES implementation and to examine the effects of different pump sizes on system performance. Furthermore, it evaluates the economic benefits of integrating PHES into the eastern Røldal-Suldal hydropower system, seeking to determine the most economically beneficial pump size.

The model used for simulating the hydropower system is Prodrisk. Prodrisk is based on a combination of stochastic dynamic algorithm (SDP) and stochastic dual dynamic algorithm (SDDP). Prodrisk is maintained and developed by SINTEF Energy research. As outlined in the Prodrisk portal, which is a platform for SINTEF researchers to extend knowledge, it is noted that "ProdRisk is a medium-term hydropower scheduling model treating inflow and power price as stochastic variables." [33]. The Stochastic Dual Dynamic Programming (SDDP) method is recognized as the leading approach for medium-term scheduling in hydropower systems [34].

2 Literature review

In this section, a short review of current research on hydropower scheduling using the SDDP method is provided, emphasizing its role in optimizing a hydropower plant across various priorities.

The SDDP method is among several methodologies developed for optimizing the scheduling of hydropower systems. In this thesis, this method has been used for hydropower scheduling of the eastern part of RSK with the implementation of PHES. The SDDP method has been implemented in several studies. For instance, in the study *Optimal Medium-Term Hydropower Scheduling Considering Energy and Reserve Capacity Markets*, by Helseth et al. [23], the SDDP method was employed to transition from solely focusing on the energy market to a combined approach that includes both energy and reserve capacity markets. The purpose of the study is to assess the expected changes in scheduling the water values when selling to both the energy and reserve markets. The study considered three scenarios: in the first scenario, the hydropower plant was optimized and scheduled solely for selling to the energy market. Scenarios two and three were optimized and scheduled for selling to both markets. These scenarios were compared. The study demonstrated that introducing a reserve capacity market is a feasible addition to an existing scheduling model.

Further on, in the article *Approximating Ramping Constraints in Hydropower Scheduling*, by Helseth et al. [35], the SDDP method is used for ramping constraints on discharge in medium-term hydropower scheduling. Similarly, in the HydroCen reports *Ramping Constraints in Hydropower Schedulling* by Helseth et al. [36], and *Environmental Constraints in Seasonal Hydropower Scheduling* by Helseth [34], both studies use SDDP-based programs aimed to address ramping constraints and environmental factors. In the first report [36], the authors aim to examine the integration of ramping limitations for particular water discharge included in the mid-term hydropower scheduling. The study shows that the current model setup for ramping constraints, which updates hourly or less frequently, overestimates how quickly water discharge can be adjusted. To address this, the paper presents a new method that calculates the costs of strictly following the rules for changing discharge rates, and it considers the financial losses from not generating power at the best prices. However, making the model more accurate in showing these changes would need a lot more computing power, which is tough since the model is already complex. The new approach has been tested and looks promising, providing results similar to a more detailed model without needing a lot more computing time. In the second report [34], the aim of the study is to identify constraints that are challenging to handle in existing models and at the same time are considered essential for users of the model to present accurately. The focus is on environmental constraints and on the SDDP method. Environmen-

tal constraints in the article refer to limits or boundaries that are put on reservoir volumes and release plans. The study identified that environmental constraints can be dealt with if state-dependent maximum discharge and virtual reservoirs can be modelled.

The SDDP method serves as a valuable tool for forecasting future scenarios and providing insights into potential outcomes. It offers hydropower producers indicators for planning and operating their schemes efficiently. These studies highlight the power of SDDP by how it can simulate scenarios with various constraints, including ramping limitations, environmental considerations and the reserve markets. These scenarios are highly relevant for both the Nordic and the European power electricity markets.

The method combines stochastic dynamic programming (SDP) and stochastic dual dynamic programming (SDDP). The model has three stochastic variables, which are the weekly inflow to each reservoir and the weekly average energy and reserve capacity prices.

Today, hydropower producers mainly obtain their income from selling energy in the day-ahead (spot) market. The active role of hydropower in balancing markets, due to its flexibility, will in turn challenge the traditional methods of operation and scheduling of hydropower systems. Deciding whether to use hydropower for electricity production or as reserve capacity affects the strategic assessment and planning of resource allocation. In the study *Optimal Medium-Term Hydropower Scheduling Considering Energy and Reserve Capacity Markets* by Helseth et al. [23], the scheduling process aims to optimize the anticipated profit from sales in both the spot and the reserve capacity markets. The purpose of the study is to measure the expected changes in scheduling and water values when selling to both markets. The method is applied in a case study of a Norwegian watercourse containing seven reservoirs with corresponding power plants. The hydropower system has a total capacity of 986 MW.

For this study, the scheduling period lasted 2 years. Scenarios for energy and reserve capacity prices were sourced from the EMPS model. The EMPS model in the study was executed using a system description of the Nordic power system, incorporating 80 years of historical inflow data, and applying reserve capacity constraints for each price zone.

Three cases were considered in the study. Case A considered the energy market only and therefore served as a reference case. Case B and Case C involve selling to both the capacity and energy markets. The scenarios vary, with Case C incorporating volume requirements. Consequently, the solution for Case C ensures that there are sufficient water levels behind the turbines to facilitate the activation of reserves.

The study concludes that the method is suitable for solving hydropower scheduling while maximizing the profit when selling to both the energy and capacity markets. The method permits the sequential allocation of resources, with the sale of reserve capacity occurring before energy sales. This study shows that introducing a reserve capacity market is feasible as an addition to an existing scheduling model. The limitation of the model is that the linearization error can have a considerable impact on the expected profits and the overall operation of some systems. Linearization is a process of simplifying complex relationships into linear ones. The error varies in the specific case. While the model is not intended to provide precise operational plans, the error from linearization might result in a misleading representation of the system's operation and profitability. The error's impact becomes more significant when considering spinning reserve capacity sales [23].

Moreover, the Nordic and European power markets are currently facing challenges related to environmental focus. Environmental regulations (obligations) within the hydropower sector are changing, focusing more on the sustainable use of water resources. The focus is to balance the use of water on hydropower, irrigation, water supply, flood control, and recreation. As a result, producers require scheduling models that accurately and consistently represent environmental constraints. In the article *Approximating Ramping Constraints in Hydropower Scheduling* by Helseth et al. [35], the authors present the modeling of ramping constraints on discharge in medium-term hydropower scheduling. Ramping constraints in the article refer to the maximum allowed changes in water flows in a hydropower system. Ramping refers to the speed at which a measured quantity varies within a physical system. The purpose of the article is to present a new technique for approximating constraints on discharge ramping by using a medium-term hydropower planning model. The used model is founded on SDDP and stochastic market prices are addressed in an outer layer through SDP. The method is a combination of SDDP and SDP programming. This model considers inflow and power prices as external stochastic variables. The model also includes ramping constraints and transition cuts (TC). With discrete time, changes in the discharge rate from one time step to another are presumed to be instantaneous. However, in reality such changes will take some time due to law of physics. Time discretization in the model enhances the apparent capability of the plant to adjust power generation in response to price changes. This means that because time is divided into intervals, the power plant's ability to quickly adapt power output in response to price fluctuations appears significantly better than it actually is. Since power price in the model is set exogenous, it allows for the calculations of costs that cover the expenses associated with changing the discharge rate. It's important to note that changes in discharge rate take time, and thus the transition costs account for these adjustments.

The model was applied to the Norwegian hydropower system Røssåga in the northern

part of the country. The hydropower system consists of two large power stations with large upstream reservoir capacity. The optimization time for the system is 156 weeks, employing 56 time steps of 3 hours each within a week. The ramping constraints is 90 m^3/s per 3-hour block. In the study, four cases were run. The cases were NoRamp, Ramp, Ramp-TC20 and Ramp-TC100. The cases NoRamp and Ramp were without and with ramping constraints respectively, and Ramp-TC20 and Ramp-TC100 were with ramping constraints and included 20 and 100 transition cost cuts respectively. The results show that the implementation of ramping constraints alone appears to slightly decrease the maximum profit in the Ramp case compared to the NoRamp case. Furthermore, the cases with TC cuts provide a notable decline in profit and an increase in computation time.

Moreover, the reservoir trajectories, which are the water levels in the reservoir, show that these were very similar for all the cases. However, in the Ramp_TC20 and Ramp_TC100 scenarios, the application of TC, leads to a more stable operation of the hydropower system due to reduced ramping frequency. There were fewer changes or variations in the water flow rates (discharge) between the minimum discharge, most efficient operation point and maximum discharge. This implies a more controlled and consistent operation of the hydropower system [35].

Although the article concludes that the study successful introduced a new method for estimating ramping constraints costs in hydropower scheduling, there are opportunity for improvement when it comes to the calculations of TC cuts. The TC cuts were determined based on the assumption that the direction of power generating ramping, either ramping up or down, is directly linked to changes in the power prices between two consecutive time periods. Accordingly, if power prices increase from one time step to the next, the model anticipates an increase in power generation. Opposite, a decrease in price leads to a decrease in generation. However, in a complex system, this straightforward relationship may not always hold true. There may be scenarios where it might be strategically or operationally beneficial to ramp up power generation when prices are decreasing or to ramp down when prices are increasing due to various reasons not directly tied to immediate price changes. That being said, the TC cuts contribute to a more stable operation of the hydropower system, which likely enhances efficiency and could lead to a better economic outcome.

Further on, the increased supply of intermittent renewable energy will give controllable generation like hydropower plants possibilities to respond to these fluctuations. At the same time, both physical and environmental demands associated with hydropower operation are changing, which restrain hydropower production. Therefore, hydropower operators must adapt their production schedules to reflect market price trends while also accommodating new operational limitations.

In the article *Environmental Constraints in Seasonal Hydropower Scheduling* carried out in a project by HydroCen [34], the author Arild Helseth addresses the importance of the need for operational scheduling models that accurately address both physical and environmental limitations. The study aimed to identify constraints that are challenging to handle in existing models and at the same time are considered essential for users of the model to present accurately. The focus is on environmental constraints and on the SDDP method. Environmental constraints in the article refer to limits or boundaries that are put on reservoir volumes and release plans. The purpose of the seasonal hydropower scheduling problem is to find the release strategies that maximize the expected profit for the period that is analyzed.

The model used in the study is the combined SDP/SDDP method. In the article, it came forward that the constraints can be addressed if state-dependent maximum discharge and virtual reservoirs can be modeled. The upper discharge boundary depends on the reservoir volume. Since the reservoir volume is state-dependent, this type of constraint is also state-dependent because it depends on the current reservoir volume. Virtual reservoirs in the article are reservoirs where the water value depends on the time it arrives or/and the path it has taken to the reservoir. Further on, Helseth also investigates how the most relevant constraints such as volume-dependent discharge boundaries and inflow-dependent discharge boundaries can be modeled in the SDDP algorithm. By modeling these constraints effectively, the study aims to improve the precision and effectiveness of hydropower scheduling tools under various state-dependent scenarios.

The study identified that environmental constraints can be dealt with if state-dependent maximum discharge and virtual reservoirs can be modelled. As already mentioned, environmental constraints in this paper refer to limits or boundaries that are put on reservoir volumes and release plans. State-dependent boundaries regulate the rate at which water discharge can be increased or decreased to protect the environment and avoid impacting downstream users. Virtual reservoirs can be set up to simulate various ramping conditions, reflecting how water from different sources or arriving at different times should be managed according to specific operational needs or environmental regulations. However, hydropower producers have not yet applied the constrained modeling to real-world scenarios, so it remains uncertain whether the proposed linear approximations are accurate enough for practical use.

In this thesis, the emphasis is on the outputs of the SDDP/SDP model rather than the model's accuracy itself. The model serves as a tool for simulating the hydropower scheme, incorporating PHES and its impact on the system. Thus, the primary focus is not on the model's precision in simulating the implementation, but it is expected to provide a reliable approximation.

3 Hydropower and Pumped Hydro Energy Storage theory

The term "hydropower" refers to the process of generating power, typically in the form of electricity, by harnessing the energy of falling water [37]. A conventional hydropower plant consist of several components such as an inlet and a dam, a waterway, a power station building, turbines, generators, a switchgear station and control systems. The intake is equipped with a trash rack that prevents large objects from entering and damaging the turbine. The waterway consists of pipes, tunnels and valves. The water makes the turbine rotate. The turbine is connected to the generator, which transforms mechanical power into electrical power. The switchgear station consists of switchgear and transformers that connect the station to the grid and transforms the voltage up to the power-grid voltage. At last, the control system includes SCADA/pls, sensors and computers [38].

3.1 Pumped Hydro Energy Storage

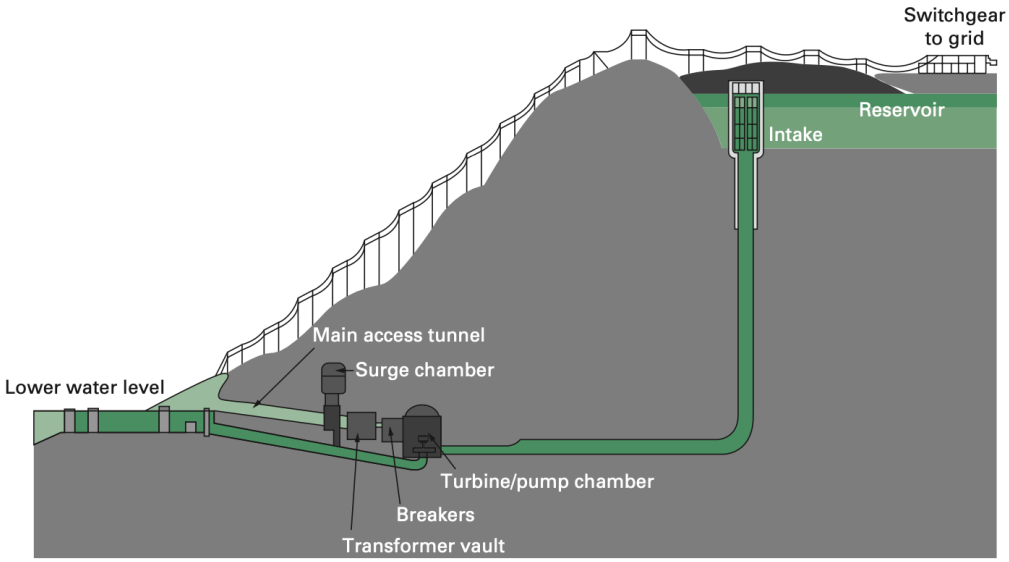


Figure 3: Illustration of a PHES inside a mountain, highlighting the system's integration with the landscape. The majority of PHES facilities in Norway are located inside mountains or underground. The figure also depicts the main components of a PHES, including the two reservoirs, intake, tunnel, turbine and pump, transformer, surge chamber, and outlet tunnel [37].

Many PHES systems are integrated with traditional river-based hydroelectric systems [4]. PHES involves accumulating water in a reservoir by pumping it uphill

during periods of low energy demand. This stored water possesses potential energy, which is harnessed and converted into electricity when released back through the hydropower plant during periods of high power demand [2]. An illustration of PHES is shown in figure 3. Moreover, there are two types of PHES: open-loop and closed-loop. Open-loop systems are defined as a plant that is permanently linked to a naturally occurring water source. A key benefit of open-loop systems is their ability to leverage existing water resources and infrastructure, reducing the need for extensive land use and construction. Closed-loop systems, on the contrary, are defined as not continuously connected to a naturally flowing water feature. A key benefit for closed loop systems is that they are less environmentally disruptive than open loop systems because they use less water and do not interfere with natural bodies of water [4].

The technical characteristics of PHES systems provide multiple benefits for the power grid. This including energy storage, enhanced grid stability, seamless integration of renewables and effective load balancing. PHES systems offer large-scale energy storage abilities. This makes PHES optimal for storing surplus power generated beyond current needs and releasing it during periods of high demand. PHES systems respond quickly to fluctuations in supply and demand, thus effectively maintaining grid stability. The capability of PHES systems to store and release energy as needed makes them highly effective at integrating renewable energy sources, such as wind and solar. These energy sources are intermittent by nature, thereby facilitating their board integration into the grid. PHES systems are excellent tools for load balancing, enhancing the power grid's overall efficiency by shifting electricity generation from periods of low demand to times of high demand. Also, PHES systems are known for long operational lifetimes and limited environmental footprint compared to other renewable energy sources. Lastly, PHES systems supply crucial ancillary services. These are, e.g., frequency regulations, voltage support, load shifting and system resilience [4].

Figure 4 illustrates a typical PHES layout with two reservoirs at varying elevations connected by tunnels or pipes. The PHES system is almost identical to a regular hydropower station, featuring the same key components. Water flows from an upper to a lower reservoir through a head race tunnel, penstock, and turbine during generation. During pumping, water is moved in reverse from the lower to the upper reservoir. Surge tanks are installed to mitigate penstock pressure fluctuations caused by 'water hammer'. Water hammer is rapid changes in pressure due to the fluid's forced cessation or change in direction. Additionally, the system includes brook intakes, also called secondary intakes, and rock or sand traps positioned before the penstock to shield mechanical components from debris. In Norway, it is common for tunnels to be D-shaped and constructed through blasting or drilling. Additionally,

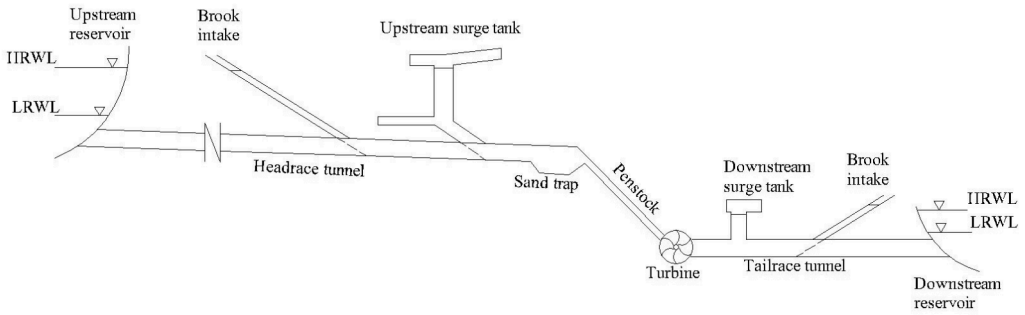


Figure 4: Layout of a Norwegian PHES. Structurally, a PHES is similar to a conventional hydropower station and includes an upstream and downstream reservoir, brook intake, head race tunnel, surge tank, pen stock, aggregate (turbine/pump and generator) and tail race tunnel. The main distinctions include the addition of a pump and motor in the aggregate for PHES operations and the need for the inlet and outlet to accommodate water flow both ways. Additionally, the surge tank is sized differently, and the placement of the sand trap is adjusted. Lastly, the power station requires greater submersion [2].

it is typical for power plants to be situated entirely underground [2].

PHES systems utilize a range of pumps and turbines to effectively store and convert energy [4]. The design of a pump-turbine primarily features two main configurations. The first and most widely employed configuration is reversible machine sets, consisting of a motor generator and a reversible pump turbine [39]. The Francis reversible pump turbine stands as the most prevalent choice [40]. Based on the rotation direction, the reversible machine functions either as a pump or a turbine [41]. The second configuration is a ternary system, which consists of a motor generator, a separate turbine and a pump. The turbine is typically Pelton or Francis. When functioning as two distinct hydraulic machines, the motor generator can maintain the same rotational direction in both modes of operation [41]. Ternary units are most commonly utilized for PHES with large head pressures, that is, between 400–600 m. Reversible turbines are used for heights lower than 400 m. The benefits of reversible turbines include their relatively compact size compared to ternary units, as well as lower investment, maintenance, and operational costs. On the other hand, ternary systems offer higher efficiency for both turbines and pumps, as each component is individually optimized for its specific function [40].

The Francis reversible turbine is designed similarly to the Francis turbine, although the runner blades are specifically designed differently. The blades must take into account both pump and turbine modes of operation. Figure 5 shows an example of a reversible turbine with key electrical components. Pressure loss in the tunnel and turbine pipes has a significant impact on the design of a reversible Francis turbine. The pressure loss reduces the gross head in turbine operation and adds to the gross

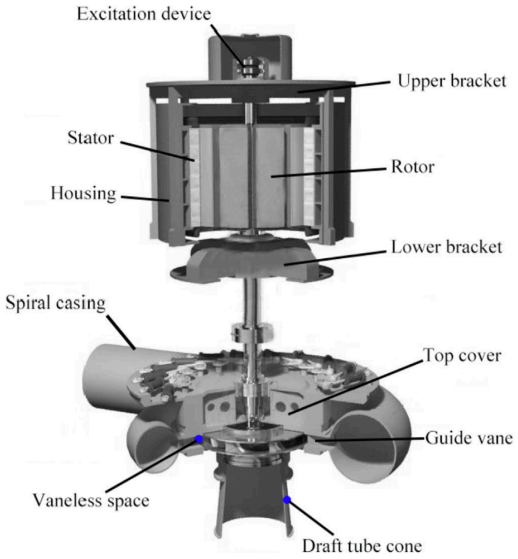


Figure 5: Diagram of a reversible pump turbine, showing key components. Water enters through the spiral casing and passes through the guide vane, which regulates flow. It then moves to the van-less space and impeller, driving the turbine's rotation, before exiting through the draft tube cone. The generator, located between the upper and lower brackets, includes the rotor and stator [42].

head in pump operation. Head is measured in meters and is the sum of the static head (the vertical difference between the two reservoirs), any losses due to friction or resistance in pipelines and equipment, and any other vertical differences that the water must overcome along the way. The net head at the turbine runner is bigger in pumping mode than in turbine mode. Achieving maximum head during the pumping mode is critical. Consequently, turbines must be designed to handle a higher pressure head than they will actually experience during turbine mode. However, this design compromise typically results in decreased turbine efficiency [40].
In a ternary system, the arrangement typically involves a Pelton or Francis turbine and a single or multi-stage pump [44]. Both the turbine and the pump are mounted on the same shaft but are separated by a clutch [45], as depicted in figure 6, with a Francis turbine. This shared shaft configuration enables both the turbine and pump to rotate in the same direction thereby facilitating faster switching between operational modes. Moreover, the independent operation of the pump and turbine enables optimization of each component, significantly improving overall efficiency [39].

Figure 7 illustrates a reversible pump turbine layout and a schematic layout of a ternary system, respectively.

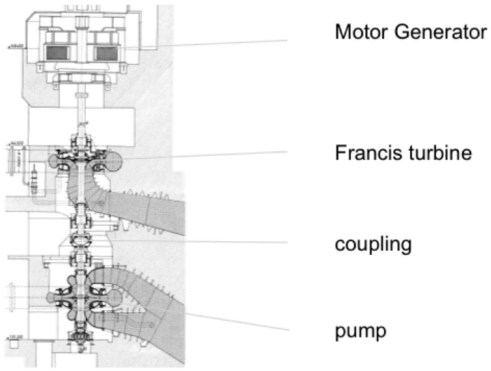


Figure 6: Diagram of a ternary pump turbine machine, featuring a motor/generator, a Francis turbine, a coupling and a multi-stage pump. All components are connected to a single shaft and rotate in the same direction [43].

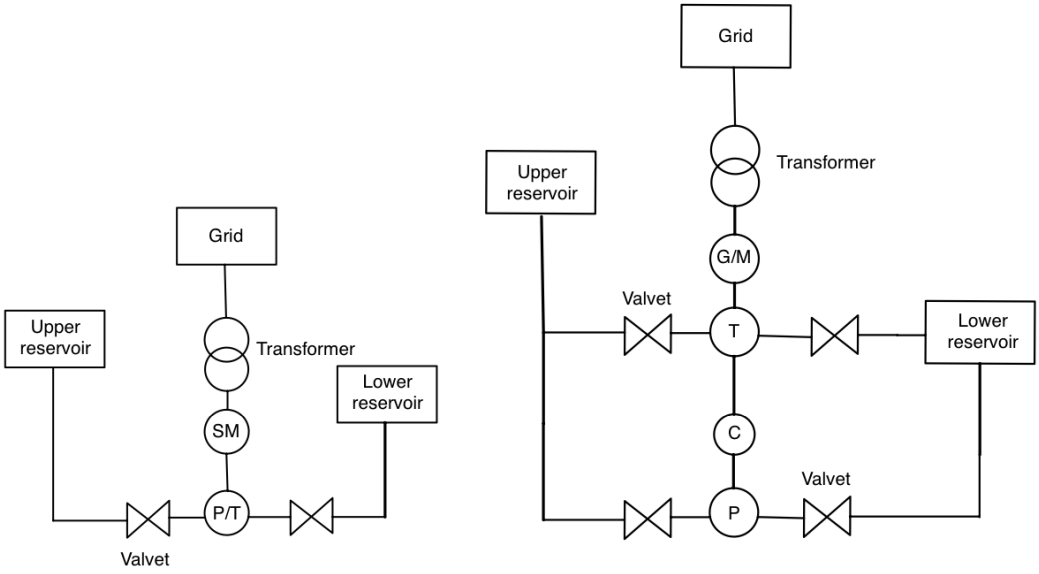


Figure 7: Layout of a PHES, with a reversible pump turbine system on the left and a ternary system on the right. In the reversible turbine system, water flows in both directions within the pipes, illustrated by black lines. In the ternary system for pumping mode, the water gets pumped from the lower reservoir through the pump and up to the upper reservoir. In turbine mode, the water flows from the upper reservoir through the turbine and down to the lower reservoir. The coupling "C" connects the turbine and pump to the generator/turbine, depending on the operation mode. Figure adapted from [45].

Each aggregate in a PHES system is equipped with key electrical components including a generator, frequency converter, high voltage switch, transformer, control

facility and a magnetization system. The generator, usually a synchronous generator, converts mechanical power to electrical power. The frequency converter facilitates startup in pump mode, enables breaking with energy recovery and controls speed in both turbine and pump operations. The high-voltage switch connects the aggregate to the grid, while the transformer adjusts voltage levels for grid compatibility. The control facility manages measurements and operation of the aggregate. Lastly, the magnetization system is crucial for both starting up and breaking the aggregate [40].

The general efficiency of turbine mode in PHES systems is approximately 90%, which indicates that 10% of the energy in the water in the upper reservoir is lost during passage through the turbine [1]. The maximum turbine efficiency for the ten PHES systems in Norway varies between 93.5% and 94%, with a round-trip efficiency ranging from 65% and 80% [2]. Furthermore, the round-trip efficiency for worldwide PHES typically falls between 70% and 85% [4] [8].

3.2 Fundamental relations in hydropower systems

The power P the turbine can generate is given by the equation (1) [46]:

$$P_{turbine} = Q \cdot \rho_{water} \cdot g \cdot H \cdot \eta_{turbinemode}. \tag{1}$$

Q represents the discharge, given in [m^3/s], which indicates the volume of water the turbine can process each second and is directly proportional to the power output. The density of water ρ_{water} is 1000 $[kg/m^3]$. The gravitational acceleration, g is 9.81 m/s^2. H refers to the head, the vertical distance the water falls. The head varies based on the water levels in the two reservoirs involved. Turbine mode efficiency $\eta_{turbinemode}$ takes into account the difference between the potential energy in the water at the upper reservoir and the electrical power output the plant deliver to the grid. The turbine mode is the operation of running water through the turbine to generate power.

The turbine mode efficiency is given by equation (2)[46]:

$$\eta_{turbinemode} = \eta_{waterway} \cdot \eta_{turbine} \cdot \eta_{gen} \cdot \eta \cdot \eta_{trans}. \tag{2}$$

The turbine mode efficiency, $\eta_{turbinemode}$ accounts for all the loss contributions. $\eta_{waterway}$ represents the energy losses attributed to the roughness of waterways, which can vary considerably. As water passes through the turbine, some of the potential energy is converted into heat. The power loss in the turbine is represented by $\eta_{turbine}$ and is between 0.8–0.95. Additionally, the generator and transformer experience minimal losses, accoun ted for by η_{gen} and η_{trans}, which typically range from 0.8 to 0.95 [37].

The power consumed by the pump or by the plant in pump mode is given by equation (3) [46]:

$$P_{pump} = \frac{Q \cdot \rho_{water} \cdot g \cdot H}{\eta_{pumpmode}}, \tag{3}$$

where Q represents the volume of water pumped from the lower reservoir to the upper reservoir, measured in [m^3/s] and correlates directly with the power consumption of the pump. H is the head in this context. It denotes the same vertical distance as in equation (1), but it measures the ascent of water during pumping. $\eta_{pumpmode}$ quantifies the efficiency loss, representing the difference between the energy used to lift the water and the potential energy stored by the water elevation. This indicates that the energy required to lift the water exceeds the potential energy gained by the water [37].

The pump mode efficiency is given by equation (4) [46]:

$$\eta_{pumpmode} = \eta_{waterway} \cdot \eta_{pump} \cdot \eta_{gen} \cdot \eta \cdot \eta_{trans}. \tag{4}$$

The pump mode efficiency also takes into account losses from the waterway, generator and transformer. The η_{pump} is the efficiency of the pump itself.

The efficiency for one cycle is given by equation (5):

$$\eta_{cycle} = \eta_{turbinemode} \cdot \eta_{pumpmode} \tag{5}$$

The efficiency of the cycle, also known as round-trip efficiency is determined by multiplying the efficiency of the turbine mode, where electricity is generated, by the pump mode efficiency, which is the process of water being pumped back to the upper reservoir [1].

Plant Kvanndal simulated in the thesis, is a combination of a turbine and a pump. These components never operate simultaneously. The overall efficiency for the plant is given by equation (6):

$$\eta_{plant} = \frac{(\eta_{turbinemode} \cdot \%t_{turbinemode}) + (\eta_{pumpmode} \cdot \%t_{pumpmode})}{\%t_{turbinemode} \cdot \%t_{pumpmode}} \tag{6}$$

The η_{plant} represents the plant efficiency over a period, which in this thesis is one year. The terms $\%t_{turbinemode}$ and $\%t_{pumpmode}$ are the percentage of time that the turbine and pump respectively are operational throughout the year.

The energy production E to a PHES station is given by equation (7) [46]:

$$E = P_{avg} \cdot t \tag{7}$$

P_{avg} is the average power output and t is the time period. The unit for energy E is given in [J].

3.2.1 Price difference factor

The price difference factor needed for making the pumping profitable is given by equation (8) [46]:

$$Price_{diff} = \frac{1}{\eta_{cycle}}. \tag{8}$$

To make the process of pumping water profitable, the selling price of the generated electricity must exceed $\frac{1}{\eta_{cycle}}$. This condition is necessary because the energy consumed to pump the water back to the upper reservoir exceeds the energy generated when the water descends through the turbines.

3.3 Stochastic Dual Dynamic Programming (SDDP)

SDDP is an optimization method used in hydropower scheduling. The strategy aims to efficiently allocate water power, accounting for the uncertainty in inflow and market price. The method aims to optimize resource allocation, thereby enhancing the potential of hydropower station planning. The SDDP algorithm employs a dynamic programming approach where future costs are modeled by hyperplanes, often referred to as cuts. This relies on linear programming. A crucial requirement is that the problem must be either linear or convex. The optimization time is divided into discrete time steps, indexed by t, and employs a finite time horizon T. A typical T for medium-term scheduling is 2 to 3 years. The time steps t are generally one week in length [5, 47]. The time optimization within a week may vary from one hour resolution to three hour resolution. Moreover, for a power station, it is assumed that its power generation P:

$$P = f(Q)h/h_0. \tag{9}$$

Function $f(Q)$ is a piecewise linear function, tailored for each power station. In the SDDP algorithm, $f(Q)$ must be concave. The variable h represents the head and h_0 is the nominal head. The SDDP algorithm cannot directly accommodate variable head values. The head correction factor h/h_0 in the equation must be employed with estimated values for h. For many Norwegian power plants, this is a reasonable approximation. This is because the head is typically quite larger relative to head variations.

The remainder of the theory underlying the SDDP algorithm is detailed in the article *An Algorithm for stochastic medium-term hydrothermal scheduling under spot price uncertainty* by Gjelsvik et al. [47] and the chapter *Long- and Medium-term Operations Planning and Stochastic Modelling in Hydro-dominated Power Systems Based on Stochastic Dual Dynamic Programming* by Gjeldsvik et al. [5]. The application of the SDDP method is described in the Methodology section 4.1.

4 Methodology

This section provides an explanation of the SDDP method used in the ProdRisk program, as well as an overview of ProdRisk itself. Additionally, the implementation of PHES is discussed. Finally, the origin of the price scenario is explained.

The program used for simulating the hydropower system is ProdRisk [33]. ProdRisk is a program for medium-term scheduling in hydropower, which considers both inflow and power price stochastic variables. The program is based on a combination of stochastic dynamic programming (SDP) and stochastic dual dynamic programming (SDDP) [33].

4.1 SDDP method

The SDDP method was developed in Brazil and is a huge breakthrough in the planning and optimization of operational strategies for hydropower systems by Pereira et al. [48]. Further on, Gjeldsvik et al. [5] emphasize in their chapter Long- and Medium-term Operations Planning and Stochastic Modelling in Hydro-dominated Power Systems Based on Stochastic Dual Dynamic Programming from Handbook of Power systems I [49]: "Using SDDP technics, however, allows stochastic optimization for multi-reservoir systems, which means that more realistic and detailed models can be dealt with."

Determining the best operational strategies for a hydropower system poses a significant challenge when dealing with a power system that heavily relies on hydropower [5]. In the Nordic power market, hydropower constitutes a substantial portion of the power production, especially in Norway where hydropower contributed 87.3% of the energy production in 2023 [9]. This makes SDDP an excellent tool for hydropower producers aiming to maximize profits in their hydropower operations [5].

SDDP is used for optimizing hydropower scheduling. SDDP is highly capable in handling the uncertainty inherent in both water inflows and to reservoirs and electricity market prices. This is very important for hydropower systems, where future water availability and market conditions can be unpredictable. The method also optimizes multi-reservoir systems. This provides a more realistic and detailed model of a hydropower system. For large-scale systems with interconnected reservoirs and power stations, such as those found in the northern market, this complexity is particularly important. The algorithm improves the decision-making by incorporating stochastic variables, inflow and price, into the scheduling process. With this, SDDP helps to make more informed and efficient decisions regarding water release from reservoirs, electricity generation, and market participation. The method is useful for medium and long-term operational planning in hydropower systems. This involves determin-

ing the optimal strategy for water reservoir management over several months to some years. It's balancing between energy production and maintaining adequate water levels.

The SDDP uses a dynamic programming technique that iteratively improves the operation strategy. Future costs are represented by 'hyperplanes' and the algorithm works through forward and backward recursion to find optimal solutions for hydropower scheduling. One complete iteration of the algorithm consists of a forward and a backward recursion [5]. As stated by the ProdRisk portal [33]: "The purpose of a forward simulation is to find a set of reservoir trajectories and compute the expected profit (lower bound). The purpose of a backward recursion is to compute cuts for each stage that describe the relationship between the future expected profit seen from that stage and the reservoir level at the end of that stage. The backward recursion also provides the future expected profit seen from the beginning of the planning horizon, which provides an upper bound. After ProdRisk has reached convergence or the maximum number of iterations has been reached, a final simulation is performed. The final simulation is a final forward simulation using the strategy (represented by cuts) created." ProdRisk simplifies the overall scheduling challenge by breaking it down into weekly decision problems. Each of these problems is structured as a problem of linear programming."

Due to the significant reliance on hydropower, market prices can be greatly influenced by hydrological conditions, making it important to consider the variability of water inflow. The model is designed for a system operated by a single power company that acts as a price taker, unable to impact the market prices. Consequently, the market price is treated as an external stochastic variable. SDDP is combined with ordinary stochastic dynamic programming to handle price modeling as an exogenous stochastic variable. A forecast of the spot price is generated through simulations conducted with the EMPS model. EMPS is a long-term model spanning multiple areas and years, using the spot price as an intrinsic variable. In the EMPS model, each price scenario is linked to a specific weather year [5].

4.2 ProdRisk and the model of Eastern RSK

In the thesis, the Prodrisk API is utilized through the 'ProdriskCore' class, which is implemented in Python. The code for this was developed in Jupyter Notebook. The primary concept of the API is to provide essential functions and simple data structures to facilitate data exchanges with ProdRisk. Its adaptability allows for easy integration with multiple programming languages. Furthermore, the object types and attributes within the API are designed to provide specific functionalities, empowering developers to perform certain operations or actions effectively using the

software's API [50].

In Prodrisk there are five primary object types that can be modified by the user. The five types are modules, pumps, inflowSeries, areas and settings. The module object represents a combination of a reservoir and a plant, featuring attributes that detail the physical characteristics of the system. These are the reservoir and plant attributes. The InflowSeries object contains water inflow data, while the area object represents the entire hydropower system. General model details are kept in the setting object, e.g., the number of SDDP iterations and price periods, to mention some. In a ProdRisk simulation, there is only one setting and one area object, but it may contain several modules, inflowSeries and pumps. Each module is required to connect to an inflowSeries using an identification number and only a single inflowSeries can be connected to multiple modules.

The model of Eastern part of Røldal Suldal hydropower system consists of four modules. The two initial modules are each composed of a single reservoir: Isvann and Holmevann, respectively. The third module integrates the Kvandandalsfoss reservoir and the Suldal II power plant, while the fourth encompasses the Sandvann reservoir and the Kvanndal power plant. This collective area is referred to as NO2. The topology is arranged such that the water follows distinct paths from Isvann and Holmevann to the Sandvann reservoir. From there, it flows to the Kvandal power station, then proceeds to the Kvandalsfoss reservoir, before passing through the Suldal II power station and ending at Suldalsvannet. Moreover, modules one, two, and four are linked to inflowSeries 1, whereas module four connects to inflowSeries 2. This implies that the Kvandalsfoss module includes extra rivers feeding into it.

Each module's reservoir is characterized by multiple attributes that provide information about it. The attributes are: max reservoir volume, an ID number which is used to connect the module to an inflow series, mean regulated inflow per year, mean unregulated inflow per year, net nominal head used, the outlet line in masl, the relation between volume in Mm^3 and head in masl of the reservoir and the start volume of the reservoir. Also, modules three and four have power plants, which are also characterized by multiple attributes that provide information about the plants. The attributes are plant name, the PQ curve (power P, discharge Q curve) which shows the relation between power and discharge, the energy equivalent constant, the maximum discharge in m^3/s, the maximum production in MW and the maximum bypass constant. All of these elements were established in the model developed by SINTEF. A modification was made in the model when implementing a pump in the system.

A pump was implemented between reservoir Sandvann which belongs to Kvanndal power station and reservoir Kvanndalsfoss which belongs to Suldal II power station.

The pump was built according to the attributes: maximum and minimum pumping height, maximum and minimum discharge, average power and pump topology in relation to the reservoirs. The position of the watercourse was at plant Kvandal, it pumped water from the reservoir Kvandalsfoss to the reservoir Sandvann. The pump is a separate pump which uses the same pipelines as the turbine. The maximum pumping height was set to the height difference between the lowest reservoir level of Kvandalsfoss and the highest reservoir level of Sandvann, and the smallest pumping height was set to the highest reservoir level of Kvandalsfoss and the lowest level of Sandvann. The discharge was calculated using equation (3), by solving it for Q. The efficiency for the pump was set to $\eta_{pump} = 0.8$. The simulation was run with different pump sizes and discharges. The pumps sizes was $P_{pump} = 10$ MW, 20 MW, 30 MW and 40 MW.

When running the model, the simulation interval has to be in whole number of weeks. For the thesis, the number of weeks was set to 208, totaling four years. The timestamp was set to 7. January 2030. Prodrisk Core does not maintain a precise correspondence with actual dates and times. The starting time is determined based on the price scenarios and the explanation is provided further down. In the simulations a three-hour time resolution is applied, whereas the actual market operates with a one-hour resolution. The input and output data have 56 three-hour time steps per week.

The number of SDDP iterations was 50 which means it was done 50 forward simulations and 50 backwards recursion. Increasing the number of iterations leads to improved accuracy in the model. Further on, the quantity of inflow scenarios and price scenarios must be the same. Each price scenario aligns with its respective inflow scenario in sequence. The number of scenarios in the thesis was 30. The time duration of each scenario needs to be at least the length of the simulation interval. This means that each inflow and price period in the simulation should cover 208 weeks. The inflow scenarios are required to span a minimum of 52 weeks.

The model ran in serial mode, that is, the simulated scenarios are constructed as sequential years. This means that the reservoir level after 52 weeks in scenario 1 becomes the initial reservoir level in scenario 2. The linking of scenarios follows a cyclic pattern. Scenario 1 initiates from the concluding state of the last scenario. In this approach, the initial reservoir level for one scenario is set based on the final reservoir level from the preceding scenario in the previous iteration. Thus, the aforementioned 30 scenarios are in serial mode and therefore 30 sequential years of simulation.
As previously mentioned, the simulation interval was for four years. When running in serial simulation, the input scenarios need to be created in accordance with the same cyclic coupling assumed by the algorithm. This means that the first scenario

consists of years 1, 2, 3 and 4; scenario 2 consists of years 2, 3, 4 and 5; the last scenario, number 30 consists of years 30, 1, 2 and 3. Results are shown for 52 weeks per scenario.

4.3 Price scenarios

The price series dataset originates from the 'Low Emission' scenario described in the report *Power Price Scenarios* by Schäffer et al. [51] from the HydroCen research center. The only difference is the entire price series was scaled up to have an average value of ca. 210 EUR/MWh, which was the average day-ahead price in NO2 in 2022.

The goal is a better understanding of price features and the factors influencing prices in 2030. In the report, two price scenarios are defined. A reference scenario and a low emission scenario for 2030. The low-emission scenario is designed to represent the latest goals and aspirations for northern Europe's power system in 2030. In the scenario, a greater portion of power production relies on wind and solar energy sources. The reference scenario is built around a renewable energy target of 27% in the energy mix by 2030. The study utilizes the EMPS model with a 2030 European dataset. This dataset encompasses 53 areas, each modeled in detail [51].

4.4 ChatGPT

ChatGPT has been utilized for spell-checking and enhancing English writing. Additionally, ChatGPT has been utilized to enhance the Python code for plotting. ChatGPT has been used in accordance with the applicable guidelines at NMBU.

5 Results

This section presents the results of the hydropower system simulations incorporating PHES. The pump sizes ranging from 0 MW to 40 MW are evaluated. The 0 MW simulation serves as the baseline and represents the system without the PHES implementation. The results are displayed in several plots, with each pump size differentiated by a unique color. The results for the reservoir levels, efficiencies, production and revenue are derived from the average values across 30 simulated scenarios for each pump size. As previously explained in the Method section, each simulation of the pump sizes comprises 30 scenarios, effectively representing a 30-year period. Additionally, the plots related to the turbine and plant, such as efficiency and duration curves, are specific to Kvanndal power station. However, the plots concerning production and revenue cover the broader area, including both Kvanndal and Suldal II power station.

5.1 Reservoir Level

The mean reservoir level for Sandvann for a year is represented in figure 8.

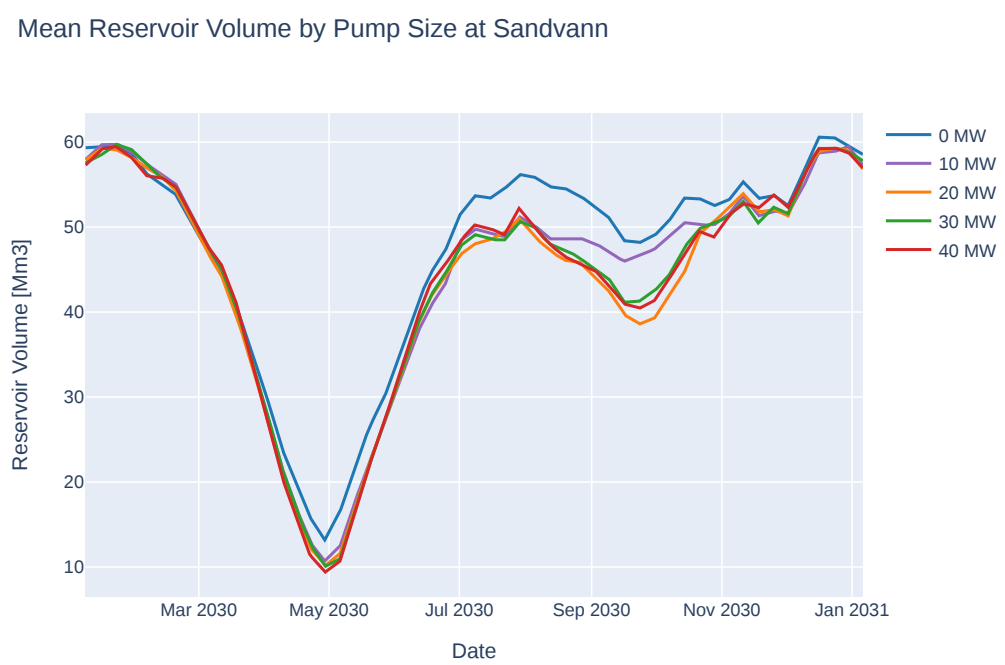


Figure 8: Reservoir level for Sandvann for various pump sizes. The y-axis indicates the reservoir level in a million cubic meters, while the x-axis tracks the dates throughout the years 2030 to 2031. Each line on the graph represents a different pump capacity, including a baseline scenario labeled as '0 MW', which shows the reservoir levels without any pump intervention.

Figure 8 shows that the Sandvann reservoir, with a maximum capacity of 66 Mm^3,

remains nearly full throughout the year for all evaluated pump sizes. It is evident that the reservoir level significantly decreases during the winter and early spring months, then rapidly increases during the flood season from May to July across all pump sizes. Additionally, the figure clearly illustrates that implementing a pump reduces the reservoir level in May for all pump sizes compared to the baseline simulation. The graphs representing the pump simulations with 10 MW, 20 MW, 30 MW, and 40 MW display minimal variation among them, with one deviation around September for the 10 MW pump graph. In contrast, the no-pump simulation consistently shows higher reservoir levels throughout the year. Overall, implementing a pump results in lower levels across all pump sizes, with the levels remaining nearly consistent regardless of the pump capacity used.

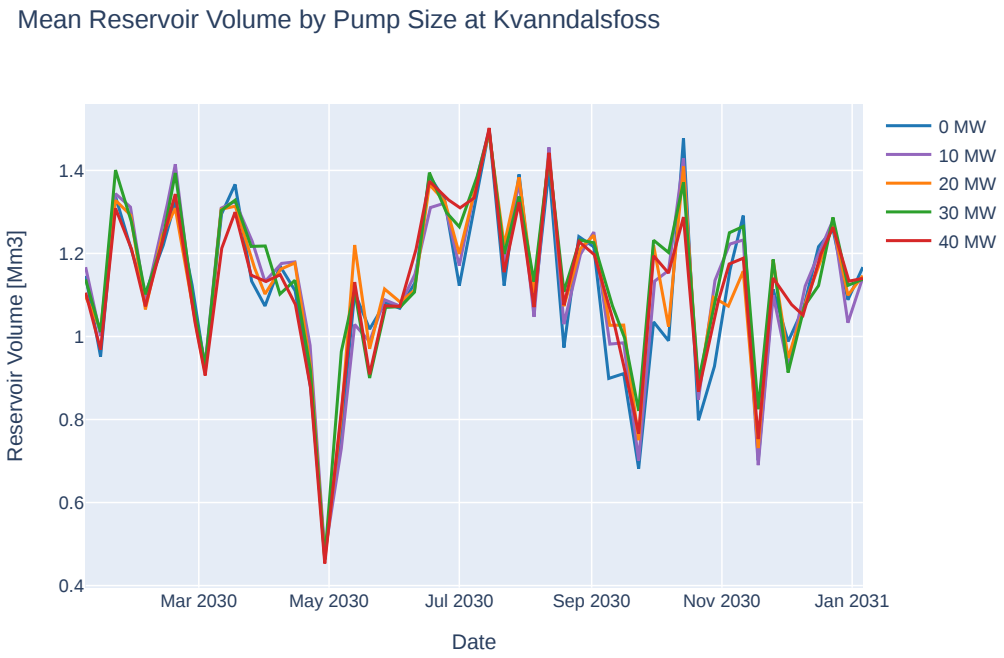


Figure 9: Reservoir level for Kvanndalsfoss for various pump sizes. The y-axis indicates the reservoir level in a million cubic meters, while the x-axis tracks the dates throughout the years 2030 to 2031. Each line on the graph represents a different pump capacity, including a baseline scenario labeled as '0 MW', which shows the reservoir levels without any pump intervention.

The mean reservoir level for Kvanndalsfoss for a year is presented in figure 9. Figure 9 shows that Kvanndalfoss reservoir, with a maximum capacity of 1.6 Mm^3, fluctuates rapidly through the year for all pump sizes. Also, for this reservoir, the graphs representing the various pump sizes display minimum variation among them. However, the baseline simulation for this reservoir does not exhibit distinct differences from the other pump simulations. Additionally, the water level in this reservoir experienced a significant drop from March to early May, followed by a rapid increase thereafter.

5.2 Duration curves

The duration curve for the turbine at plant Kvanndal is presented in figure 10.

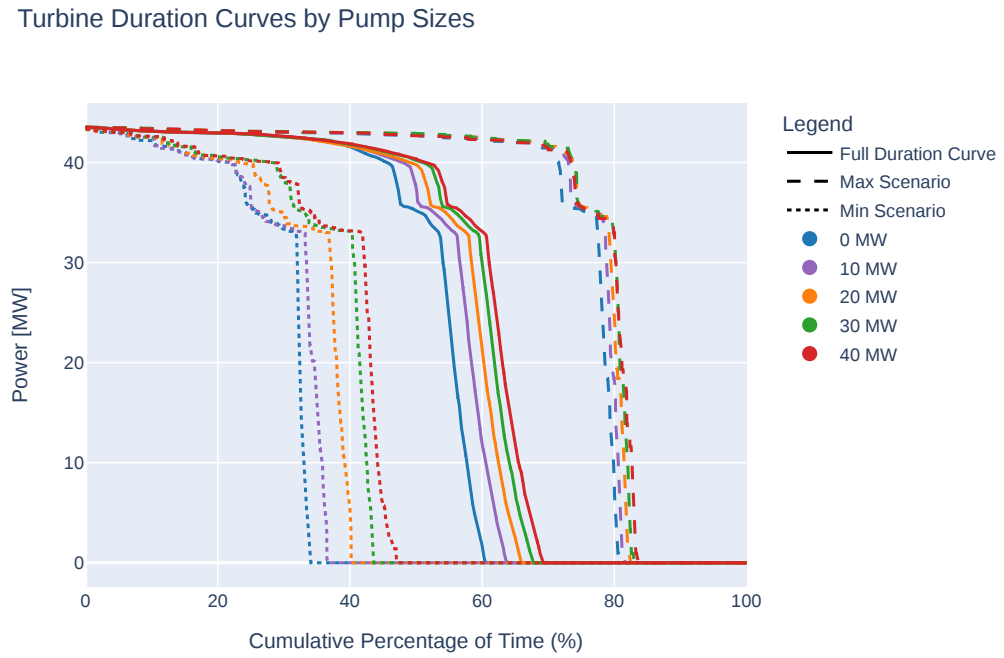


Figure 10: Duration curves for the turbine at plant Kvanndal for various pump sizes. The three duration curves represent data from the year with minimum power output, an aggregate 30-year period, and the year with maximum power output. The y-axis indicate the power output, given in million of watts, while the x-axis gives the total percentage of time for a year.

The duration curve illustrates the percentage of time the turbine operates at various power levels. The figure shows only the power output from the turbine. Figure 10 displays three distinct plots. Each solid line represents the duration curve for the turbine for each pump size for the 30-year period. The dotted lines on the left side of the plot indicate the years with the lowest mean power output, while the dashed lines on the right side represent the years with the highest mean power output. Across all three plots, the figure demonstrates that larger pumps enable the turbine to operate over an extended period. Additionally, in all three plots, the red lines representing simulations with the 40 MW pump show a more gradual decline in power across the cumulative percentage of time than the lines representing other pump sizes. This shows that the turbine with 40 MW pump has a longer operating duration than it would have with smaller pump sizes. The operating duration varies significantly more during years with the lowest mean power output compared to the overall 30-year period and the years with the highest mean power output. On the

other hand, during years with the highest mean power output, the differences in operating duration across various pump sizes are notably smaller.

The duration curve for the plant Kvanndal is presented in figure 11. The duration

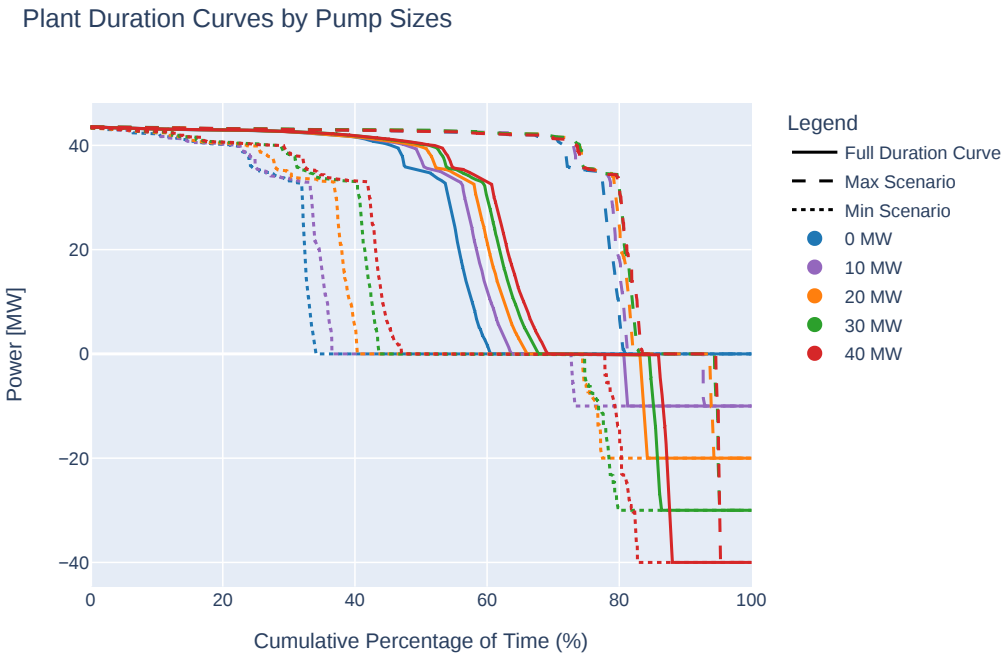


Figure 11: Duration curves for plant Kvanndal for various pump sizes. The three duration curves represent data from a year with minimum power output, an aggregate 30-year period, and a year with maximum power output.The y-axis indicate the power output, given in million of watts, while the x-axis gives the total percentage of time for a year.

curve for the plant combines the turbine's duration curve for various pump sizes, as shown in figure 10, with the duration curve for different pump sizes in figure A.3, which is found in the Appendix. The three distinct plots represent the same as in figure 10. Negative values on the curve indicate the pump's operation times, which are considerably shorter than those of the turbine. This suggests that the power plant operates most of the time in turbine mode. Additionally, during years with the lowest mean power output, the pump's operating duration shows greater variability compared to the overall 30-year period than the years with the highest mean power output. On the other hand, in years with the highest mean power output, the operating duration for the plant remains almost consistent across all pump sizes.

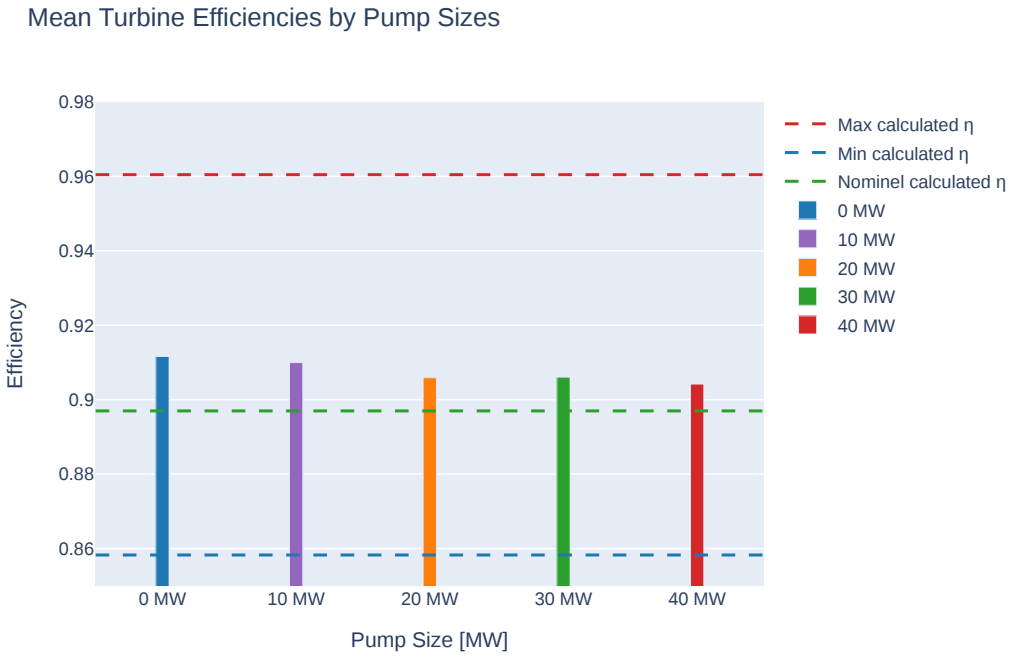


Figure 12: Mean turbine efficiencies for various pump sizes and the theoretical maximum, minimum and nominal mean efficiencies. The y-axis indicate the efficiency, while the x-axis gives the pump sizes.

5.3 Efficiencies

The mean efficiencies for the turbine for different pump sizes are presented in figure 12. The efficiencies for the turbine vary very little for the different pump sizes. The turbine efficiencies, $\eta_{turbinemode}$ are 0.9116, 0.9099, 0.9059, 0.9060 and 0.9042 for the pump sizes 0 MW, 10 MW, 20 MW, 30 MW and 40 MW, respectively. The efficiencies decline with increasing pump size. The dotted lines are derived from equation (1) and the PQ curve, which charts the relationship between power output and discharge. The red dotted line represents the maximum theoretical efficiency of the turbine. This is calculated using corresponding P and Q from the PQ curve, the greatest possible height difference between the Sandvann and Kvanndalsfoss reservoirs, and then taking the mean of these calculations. The blue dotted line indicates the minimum theoretical efficiency achievable by the turbine. It is calculated similarly to the maximum efficiency but uses the smallest height difference between the Sandvann and Kvanndalsfoss reservoirs. The green dotted line illustrates the nominal theoretical efficiency of the turbine. It is calculated using the same method as the previous two, utilizing instead the nominal height difference between the Sandvann and Kvanndalsfoss reservoirs.

Table 1 displays the annual percentage of time that the turbine and pump are running through the year for each pump size.

Table 1: Operational time for turbine and pump for various pump sizes

Pump Size (MW)	**Turbine Running Time (%)**	**Pump Running Time (%)**	**No Running Time (%)**
0	60.53	0.00	39.47
10	63.74	19.32	16.94
20	65.97	16.87	17.16
30	67.68	15.48	16.84
40	69.15	14.06	16.78

Table 1 indicates that as the pump size increases, the turbine's running time also increases while the pump's running time decreases. Additionally, the amount of time the plant is inactive decreases as the pump size increases.

The mean efficiencies for the plant for different pump sizes are presented in figure 13. The efficiencies of the plant marginally increase as the pump size grows. No-

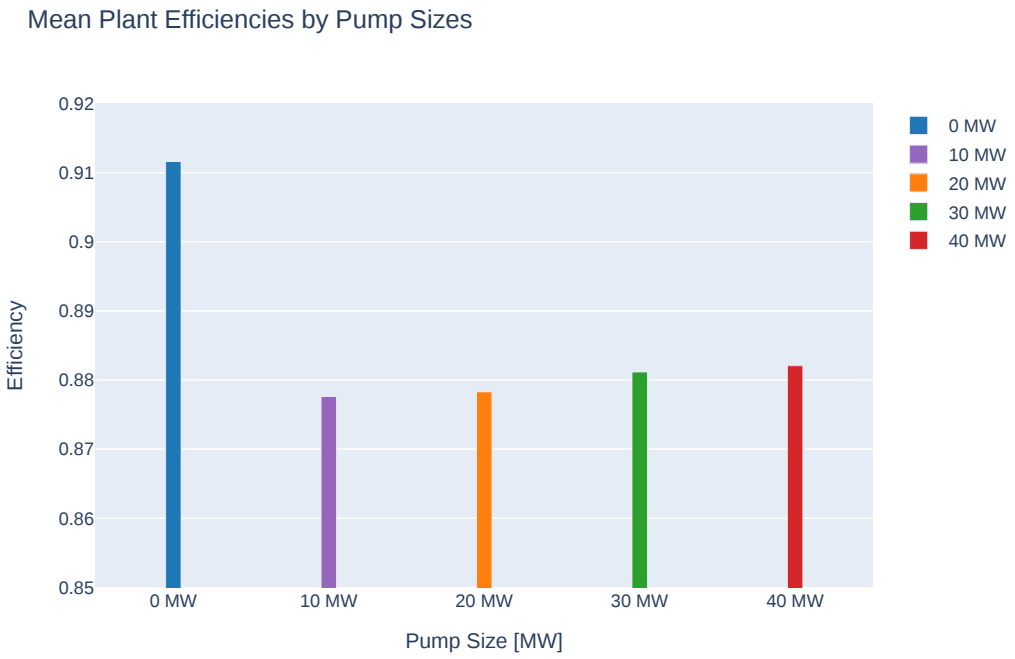


Figure 13: Mean plant efficiencies for various pump sizes. The y-axis indicate the efficiency, while the x-axis gives the pump sizes

tably, the plant operates at its highest efficiency, η_{plant} at 0.911 without a pump. For increasing pump size of 10 MW, 20 MW, 30 MW and 40 MW the efficiencies, η_{plant} are 0.8776, 0.8783, 0.8811 and 0.8820 respectively. These plant efficiencies are calculated using equation (6), incorporating the turbine efficiencies presented in figure 12, the pump efficiencies uniformly set at 0.8 and the percentage of turbine and

pump running times in table 1.

5.4 Production

The mean annual production of the two power plants, Kvanndal and Suldal II for various pump sizes is depicted in figure 14.

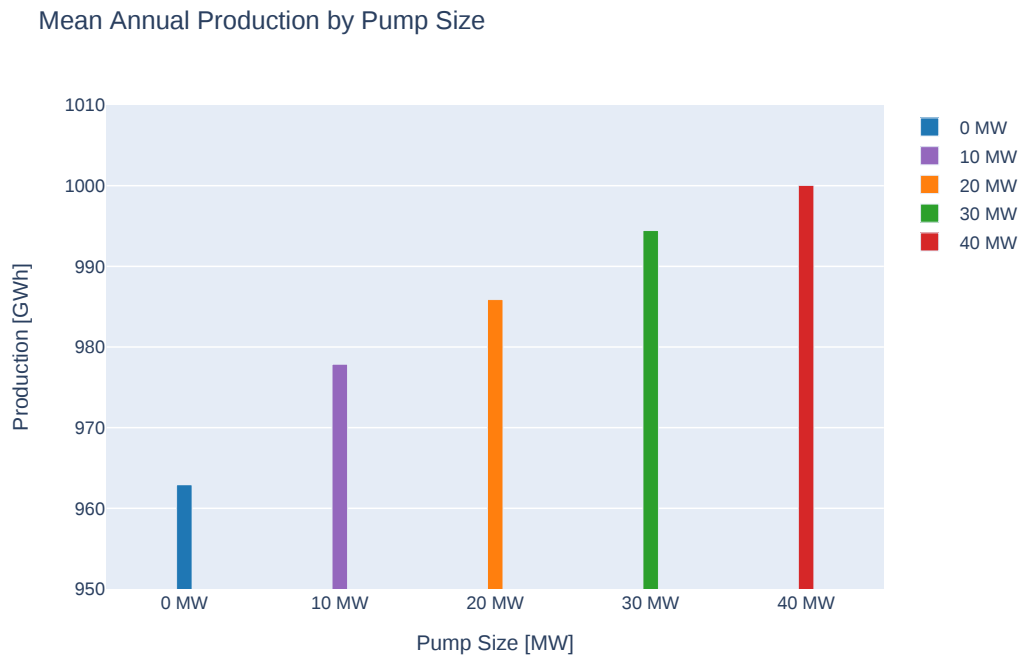


Figure 14: Mean annual production from Kvanndal and Suldal II for various pump sizes. The y-axis indicate the mean annual energy production in gigawatt-hours, while the x-axis gives the pump sizes.

The figure indicates that the mean annual production increases with an increasing pump size. The mean annual productions are 962.92 GWh, 977.86 GWh, 985.88 GWh, 944.44 GWh and 1000.03 GWh for the pump sizes 0 MW, 10 MW, 20 MW, 30 MW and 40 MW respectively. The production percentages relative to the baseline simulation are illustrated in figure A.5 in the Appendix. Compared to a baseline simulation, the production increases are 1.51% for 10 MW pump size, 2.31% for 20 MW, 3.18% for 30 MW and 3.74% for 40 MW. The mean annual net production of the two power plants, Kvanndal and Suldal II for various pump sizes is depicted in figure 15.

The mean annual net production is calculated by subtracting the mean annual energy used for pumping, as shown in figure A.6 in the Appendix, from the mean annual production figures presented in figure 14. The net productions for pump sizes of 0 MW, 10 MW, 20 MW, 30 MW, and 40 MW are 962.92 GWh, 961.18 GWh, 957.29

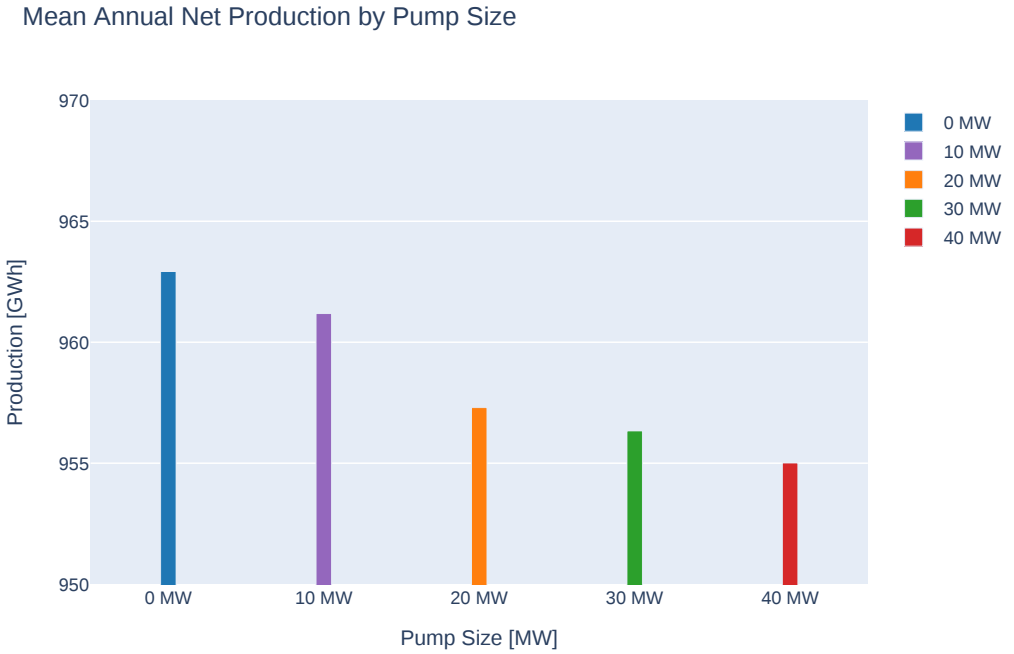


Figure 15: The mean annual net production from Kvanndal and Suldal II for various pump sizes.The y-axis indicate the mean annual net energy production in gigawatt-hours. This is the generated energy from the plant over the course of a year, subtracted by the energy required for pumping operations throughout the same, time-frame. The x-axis gives the pump sizes.

GWh, 956.33 GWh, and 955.01 GWh, respectively. Correspondingly, the energy required for pumping, recorded as 0, −16.76 GWh, −28.58 GWh, −38.10 GWh, and −45.02 GWh for each size, increases with pump size. This data illustrates that larger pumps lead to higher annual energy consumption due to increased pumping requirements.

5.5 Revenue

The mean annual revenue in percentage relative to the baseline simulation with no pump, for various pump sizes, is presented in figure 16.

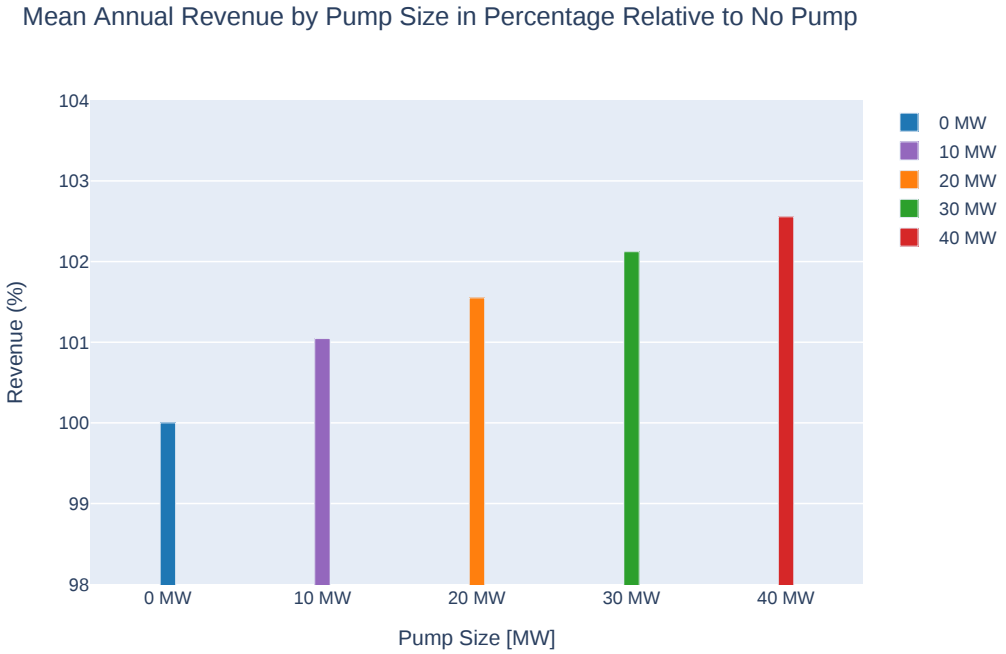


Figure 16: Mean annual revenue in percentage relative to no pump for various pump sizes. The y-axis represents the revenue for the power stations Kvanndal and Suldal II, given in percentages. Here, the income from the baseline simulation, is set to 100%. The revenue for simulations with different pump sizes is presented as a percentage compared to this baseline. The x-axis displays the different pump sizes.

The figure illustrates that annual revenue increases with larger pump sizes. The revenue represents the profit that hydropower producers retain after a year of selling and purchasing electricity. It accounts for the costs associated with buying electricity for pumping operations. Revenue is set to 100% for the baseline simulation. Implementing a 10 MW pump results in a 1.05% increase in annual revenue. For pump sizes of 20 MW, 30 MW and 40 MW, the revenue increases by 1.55%, 2.12% and 2.56%. Figure A.11 in the Appendix presents the revenue in millions of euros.

6 Discussion

The research questions explored in this thesis were *How does employing PHES with the SDDP method impact the eastern Randal-Suldal hydropower systems operations? How do pump size variations affect the Kvanndal power plant's efficiency and performance?* Consequently, this study delves into the operational effects of PHES on the hydropower system, specifically examining the implications of utilizing pumps of varying sizes. Additionally, it explores the economic impacts of integrating PHES and identifies the most financially viable solutions for implementation

In the thesis, a PHES was placed at the Kvanndal power station, utilizing Sandvann as the upstream reservoir and Kvanndalfoss as the downstream reservoir. One of the reasons for positioning the pump at this location is to make use of the water from Kvanndalfoss. Kvanndalfoss is significantly smaller reservoir, with 1.6 Mm^3, in contrast to Sandvann's 66 Mm^3 and Holmevann's 96 Mm^3. Kvanndalfoss receives water from Kvanndal and adjacent streams. During flood season, a considerable amount of water is spilled as the reservoir lacks the capacity to contain all the floodwater.

In March 2024, Lyse DA submitted a concession application for the Kvanndal 2 PHES. In the application, it is noted that the significant unregulated inflow in Kvandalsfoss leads to considerable water and potential energy losses in the eastern part of RSK. During the flood season, the Suldal power station is forced to operate at low power prices to prevent water loss. Furthermore, the concession application reveals that a primary motivation for constructing a PHES is the growing frequency of hours when energy prices are near zero or negative. These negative energy prices signify a surplus of production, which in turn yields low societal benefit. The application further highlights the potential advantages of PHES. With PHES in the system, it enables the storage of water in reservoirs for use during periods of high consumption and low output from solar and wind power sources. Thus, PHES has the capability to transform energy with low societal benefit into energy with greater societal value. Based on the motivation, it can be inferred that economic considerations are the primary drivers behind the concession application. Also, in general, profitability largely dictates whether a hydropower producer should consider expanding or modify production. This is also evident in the master thesis, where the SDDP method has been utilized for optimization based on revenue. From the optimization, the primary focus has been to identify which pump size generates the maximum revenue.

The concession application confirms that PHES could be profitable in the eastern part of RSK. However, the scenario presented in the application differs from that in the thesis. Lyse DA intends to use the Kvanndalsfoss reservoir as the downstream reservoir and Holmevann as the upstream reservoir. Holmevann has a regulative height of 10 m from 958 masl to 948 masl, but in the concession they want to ex-

pand this with an additional 5 m. The choice of placement for the pump appears wise, given the substantial reservoir capacity of Holmevann. In this thesis, one of the primary emphasis was examining how implementing a pump to the existing Kvanndal power station impacts the power system rather than integrating a new power plant with PHES. That being said, Sandvann also possesses a significant regulatory reservoir capacity as it can regulate between elevations of 950 masl and 924 masl, which is a 26 m range. This substantial regulatory height provides the reservoir with good conditions to function as an upper reservoir for a PHES.

The reservoir level of Sandvann in figure 8 shows that implementing a pump affects the reservoir level of Sandvann. There is minimal variation in the reservoir level across different pump sizes. The line representing the baseline simulation is distinct from the others. The figure illustrates that implementing a pump results in a lower reservoir level compared to the simulation without a pump. During summer months, simulations with pumps remain lower than the simulation without. This is particularly evident in the weeks leading up to the flood season, as demonstrated by the significant 'dip' observed in the figure. The figure suggests that using a pump provides hydropower producers the flexibility to use water and generate electricity, as the pump enables them to restore the reservoir volume. Without a pump, producers must manage water usage conservatively and exercising caution in how they utilize the limited water available in the reservoir.

Continuing with the reservoir level at Kvandalsfoss, which is the downstream reservoir for the PHES. There is no significant variation in reservoir level across different pump sizes, as observed in figure 9. Throughout the year, the reservoir level fluctuates rapidly. There is also a marked 'dip' in the weeks leading up to the flood season, which aligns with the dip observed at Sandvann. Kvanndalfoss is significantly smaller with its 1.6 Mm^3 compared to Sandvann with 66 Mm^3. The small reservoir not only gets water from Kvanndal, but also from other streams. The fluctuations should be interpreted as a result of the reservoir's limited capacity. For instance, when pumping or high generation occurs in Suldal and a significant amount of water is removed, the reservoir level drops drastically. Conversely, when there is a high inflow, the reservoir fills quickly.

Continuing the analysis and examining the plant duration curve for Kvanndal as shown in figure 11, it becomes evident that implementing a pump significantly affects the turbine's operating time. Specifically, observing the dotted lines, it is apparent that having a pump during years of low power output increases the pump's operating time. The low power output years should be interpreted as years with low inflow. For instance, the red dotted line representing the 40 MW pump shows the turbine operating 50% of the time during the year, whereas the blue dotted line representing the baseline simulation which is without a pump shows that the turbine operates

only 35% of the time. This difference is significant. Furthermore, the red dotted line representing the 40 MW pump displays a steady decline in the graph from 40 MW down to 30 MW, indicating that the turbine operates for a longer duration at higher outputs compared to smaller pump sizes. At 40 MW power output, the red dotted line shows a 30% operating time, while the blue dotted line is around 22%. Additionally, from ca. 34 MW to 40 MW the operation time for the red dotted line decreases from 41% to 30%. In contrast, the blue dotted lines operation time drops from 30% to 22% in the same range. However, when examining the dashed lines on the left, which represent years of high power output and consequently high water flow. The pump has a minimal effect on the turbine operation time, as both lines almost intersect at zero on the x-axis. This suggests that during years with high water inflow, the necessity for a pump diminishes as reservoirs consistently remain at full capacity. When examining the dotted lines for the pump, which appear in the part of the graph below zero, it becomes evident that larger pump sizes operate for longer duration during these years. For the dashed lines which represent the years with the highest mean power output, there is almost no difference between the pump sizes.

The solid lines, previously mentioned in the results section, represent the duration curve for the power plant over a 30-year period. These lines incorporate all 30 scenarios, which are sequential years. These lines also include the years with the highest and lowest inflow. The solid line illustrates what can typically be expected over such a period. This makes the solid lines more representative of the plant's operational duration. The differences between these lines are not as marked as in years with the lowest inflow. However, there is still considerable variation in operation times for different pump sizes. Specifically, the red line representing the 40 MW pump shows that the turbine operates more compared to the turbines in the other simulations. Figure 11 clearly illustrates that the simulation with a 40 MW pump, operates at 40 MW power output 50% of the time, whereas the simulation with no pump operates at 45% for the same power output.

Moving on to the turbine efficiencies for different pump sizes. In figure 12 the turbine efficiency $\eta_{turbinemode}$ decreases with the increasing pump size. The highest turbine efficiency occurs in the baseline simulation, where the efficiency is at 0.911. Yet, the differences between sizes are minimal. The difference in turbine efficiency between baseline simulation and with a 10 MW pump is only 0.5%. Between a 30 MW and a 40 MW pump, the efficiency difference narrows further to just 0.1%. The simulation with a 40 MW pump exhibits the lowest turbine efficiency.

In examining equation (1) to solve for $\eta_{turbinemode}$, it is observed that power P is directly proportional to efficiency. The variable H, representing the head, fluctuates with the reservoir levels between the two reservoirs. The maximum head is when

the upper reservoir is at its highest and the lower at its minimum. A larger head allows water to fall from a greater height, thus generating more power. Observations of Sandvann reservoir levels in figure 8 reveal similarities between reservoir levels and efficiencies. The levels are lower when pumps are operational, and these lower levels correspond to simulations with reduced efficiencies. This implies that in the simulation without the pump, reservoir levels are consistently higher throughout the year, resulting in a greater head and higher turbine efficiency. Conversely, simulations with pumps exhibit lower reservoir levels, reducing the head. These simulations demonstrate lower turbine efficiency.

Furthermore, the duration curve for the turbine, shown in figure 10, indicates that the turbine achieves higher power output over a longer duration with increased pump size. This increase in power output correlates with a higher discharge rate. However, an examination of the PQ (pressure-flow) curve for the turbine reveals a distinct pattern. The PQ curve plots the relationship between the pressure and the flow rate of the turbine and is shown in figure A.12 in the Appendix A.1. The PQ curve is important for understanding the turbine's performance under different operation conditions. The PQ curve for the turbine from a relatively flat curve. However, the curve shows a slightly steeper slope for lower values of P and Q. This pattern indicates that the highest efficiencies are achieved at these lower levels. When pumps are implemented, although the power output and discharge rate increase, this shift might move the operating point to a less efficient region of the PQ curve. Additionally, as the head H increase, it causes the PQ curve to shift upward, reflecting an increase in power output P. This demonstrates that a higher head contributes to enhance efficiency.

Moving on to the plant efficiency illustrated in figure 13, the most notable simulation is the baseline simulation, achieving an efficiency of 0.911. According to the equation (6), the plant efficiency η_{plant} is determined by dividing the product of the pump's and turbine's efficiencies by the sum of their total operational times. In the baseline simulation, the plant's efficiency is equivalent to that of the turbine.

The figure also shows that plant efficiency progressively improves with each increase in pump size, beginning at 10 MW, however the differences in efficiencies are minimal. Specifically, each increase in pump size results in a marginal efficiency gain of 0.01. It can be confidently stated that the overall efficiency of the plant stabilizes around 0.88 for all pump-size simulations. Although the pump efficiency was initially set at 0.8 across all sizes, figure A.4 suggests a decrease to approximately 0.77. As previously explained, the plant efficiency is derived from the equation (6) which incorporates both turbine $\eta_{turbinemode}$ and pump efficiencies $\eta_{pumpmode}$, along with operational time from table 1. Since the turbine operates for a significantly longer duration than the pump, its efficiency has a greater influence on the overall plant

efficiency compared to the efficiency of the pump. Therefore, these plant efficiencies are consistent and logical given the provided data. Additionally, the overall plant efficiency increases with larger pump sizes, contrasting with the turbine efficiency, which decreases as pump size increases. This may be attributed to the pump efficiencies detailed in figure A.4, which are 0.7709, 0.7703, 0.7723, and 0.7732 for pump sizes of 10 MW, 20 MW, 30 MW, and 40 MW, respectively. These pump efficiencies play a crucial role in the computation of overall plant efficiency. If the pump efficiencies were consistently set to 0.8, the overall plant efficiency would mirror the pattern observed in the turbine efficiency.

In summary, turbine efficiency across various pump sizes is approximately 0.91, while plant efficiency is around 0.88. Referring back to the theory section 3.1, for Norwegian PHES systems, turbines have a maximum efficiency of about 94% [2]. Additionally, the general efficiency in turbine mode for PHES systems is around 90% [1]. These findings are closely aligned with the results presented about the turbine efficiency, confirming the theoretical expectations. Which further suggests that the power plants's efficiency is reasonably accurate.
When it comes to production, figure 14 shows that yearly output increases with the size of the pump. Additionally, as illustrated in figure A.5 in the appendix, the simulation with the largest pump of 40 MW indicates that the turbine generates 3.74% more energy throughout the year compared to the setup without a pump. Moreover, other simulations with pump sizes ranging from 10 MW to 30 MW also demonstrate increased production, enhancing yearly output by 1.51% to 3.18%. At first glance, the percentage increase between various pump sizes seems minimal. However, since energy production is measured in gigawatt-hours, an increase of 3.74% represents a substantial production boost. Implementing a 40 MW pump enables the turbine to generate approximately 1000 GWh compared to 962.9 GWh without a pump. This increase of about 37.1 GWh occurs specifically when there is a demand for energy, enhancing the system's responsiveness to peak consumption times. Considering, the average annual energy consumption of a Norwegian household ranges from, 20000 to 25000 kWh. This increase in production could supply energy to between 1484 and 1855 households annually.

On the other hand, figure 15 clearly shows that net annual production decreases as pump size increases. This trend is explained in figure A.6, which details the yearly energy consumption for pumping. For the simulation involving the largest pump, the consumption reaches 45 GWh. This consumption leads to an increase in production of 37.1 GWh, revealing that the pump consumes more energy than the turbine generates. Specifically, from the 45 GWh required to pump a certain volume of water, only 37.1 GWh of energy is recovered, equating to approximately 82.4% efficiency. This value shows a moderate consistency with the efficiency of the plant with the 40 MW pump, which is 88.2%. However, 88.2% is the mean annually plant efficiency,

which is calculated from equation (6). The 82.4% is derived from the enhanced energy production attributable from the pump. This calculation considers only the pumped water and the energy derived from it, without accounting for the various production stages throughout the year. It is the cycle efficiency. However, applying equation (5) for the 40 MW simulation, where the turbine efficiency $\eta_{turbinemode}$ is 0.904 and the pump efficiency $\eta_{pumpmode}$ is 0.773. This yield a cycle efficiency at 0.698. This indicates that the cycle efficiency derived from the output is higher than the plant efficiency calculated using equation (5). The turbine have higher efficiency than the pump. Therefore, could this be due to the turbine operating for longer periods than the pump. This could be attributed to the pump increasing the capacity in the lower reservoir at Kvandalsfoss, allowing the turbines to operate for extended periods. Referring back to the theory section 3.1, several studies emphasize that overall plant efficiency typically ranges from 70% to 80% [4, 8]. Additionally, for Norwegian PHES systems, have an overall plant efficiency ranging from 65% to 80% [2]. These findings indicate that the plant cycle efficiency of 82.4% is marginally higher than expected and underscores that turbine may operating for longer periods than the pump. Further on, a significant question remains: Is the trade-off using 45 GWh to generate 37.1 GWh of electricity profitable? This question might be answered by examining the revenue figure provided later in the document.

Moreover, according to table 1, it is evident that the turbine with larger pump implementations operates for longer periods. Similarly, the larger pumps run longer compared to smaller pumps. This observation aligns with expectations, as a larger pump consumes more power, which in turn increases the volume of water pumped back into the reservoir. Consequently, with more water available, the turbine operates under increased load to generate more energy, leading to a rise in discharge rates. Therefore, with an enhanced water supply, both the turbine and pump experience extended operational periods. Consequently, a larger pump not only extends the operational periods of both the turbine and pump but also enhances energy production through greater pump capacity and the necessary handling of increased water volumes. Despite the lower efficiency observed in the simulation with the highest pump capacity, the turbine produces more energy annually when equipped with the 40 MW pump compared to the other simulations.

The final aspect of the results concerns revenue. As previously mentioned, profitability is a key determinant in whether a hydropower producer should consider expanding to modify production. As stated earlier, the main focus of this thesis has been to identify the pump size that is most economically viable. In this thesis, revenue refers to the income that power producers earn from selling generated electricity to the power market; it is important to note that this does not represent the profit itself.

Figure 16 illustrates that annual revenue increases with the size of the pump. This

indicates that although the net annual production decreases as the pump size increases, the additional production still yields more income compared to the baseline simulation without a pump. For instance, in the simulation with the 40 MW pump, which consumes 45 GWh annually, there is an increase in power production of 37.1 GWh compared to the simulation without the pump. This additional 37.1 GWh results in a revenue increase of 2.56%. Since the revenue is given in millions of euros, an increase of 2.56% represents a substantial boost in revenue. However, this thesis does not account for the investment costs associated with implementing a pump, which could significantly impact the analysis of annual revenue. Also, the payback period significantly influences the profitability of the investment.

Moreover, according to equation (8), the price difference factor for the simulation with the 40 MW pump is 1.43. This indicates that the price differential between electricity consumption and production must be 43%. Therefore, to achieve profitability, the prices during electricity generation need to be at least 43% higher than during pumping.

In this thesis, the ProdRisk program was utilized, which employs the SDDP algorithm. This method optimizes and schedules power production and revenue by taking into account stochastic variables such as inflow and power prices. The optimization is based on the data sets of price scenarios and inflow patterns. The dataset detailing inflow comes from NVE's database and represents local inflow data spanning the area from 1981 to 2010. This 30-year period encompasses a wide variety of inflow conditions, likely reflecting a significant range of variability across different years.

The power price dataset used comes from the HydroCen study *Power Price Scenarios* by Schäffer et al. [51]. As noted in the method section 4.3, this dataset specifically draws from the Low Emission scenario, which aligns with the latest climate targets for Northern Europe set for 2030. This scenario anticipates a significant increase in power production from wind and solar sources. In contrast, the reference scenario is based on the EU's target of achieving 27% renewable energy in the European energy mix by 2030.

The study utilized the EMPS model, analyzing a dataset projected for Europe in 2030. This dataset encompasses 53 detailed modeled areas and three exchange connections, with simulations spanning 58 years and capturing 56 three-hour time steps per week. The analysis includes calculations of water value based on 58 historical weather years. According to the report, the integration of renewables in the Low Emission scenario contributes to increased unpredictability and larger price fluctuations. This impacts the power prices significantly.

The EMPS model is widely used for forecasting future spot prices in the Nordic

power systems, and has also garnered interest in Asia. However, this market model primarily relies on historical data to make predictions, which limits its ability to forecast extreme price variations. Consequently, the model tends to underestimate the fluctuations in extreme prices, a significant limitation since these extremes are challenging to predict. This underestimation could lead to speculations that power producers might actually expect higher revenues than the model suggest.

The price series in the dataset has been scaled up to have an average value of 210 EUR/MWh. This is the average day-ahead price in 2022 for zone NO2. This is considerably higher compared to the average of both previous and future years [52]. Additionally, the price scenario article predicts that the average power price in 2030 for the Low Emission scenario in southern Norway will be significantly lower at 39 EUR/MWh. This discrepancy suggests that the price data in the dataset has been scaled up substantially, maintaining the same fluctuations but at an elevated level. Consequently, this adjustment may lead to expectations of higher revenue than what might realistically occur. The values depicted in figure A.11 in the appendix indicate unrealistically high revenue. However, figure 16 compares these values relative to the no pump simulation, showing that the proportions remain consistent even if the price series were scaled up to a higher average price.

Further examination of the high revenues depicted in figure A.11 in the appendix reveals that the revenue is approximately 239 million euros without the pump and around 245 million euros with a 40 MW pump. Despite these unusually high numbers, they do not conclusively determine the profitability of investing in a pump or the optimal size of the pump, for several reasons: Firstly, ProdRisk does not account for maintenance costs, which were not included in the simulations. Had maintenance been considered, which implies operational pauses, the projected revenue would likely be lower. Additionally, when contemplating the investments in a pump, producers must consider associated costs such as new tunnels, downtime at the power station and other expenses related to the investment in PHES. While investing in a pump could potentially increase annual electricity revenue, there is no guarantee that such an investment would be profitable.

Moreover, considering revenue, in 2022 the effective ground rent tax was set to 45%, resulting in an overall marginal tax rate of 67%. Consequently, power producers exceeding a certain size are required to pay 67% of their profits in taxes [53]. This is a significant factor, as it substantially reduces the post-tax earnings of hydropower producers. Additionally, between 2022 and 2023, the government introduced a high-price contribution for hydropower, setting the duty at 23% for any monthly price that exceeded NOK 70 øre/kWh. This meant that when spot prices surpassed 70 øre/kWh, producers were obligated to pay 23% of their revenue generated above this threshold [54]. Although the government decided to abolish the high-price con-

tributions in late 2023, it remains important to recognize that the exploitation of Norway's natural resources is subject to heavy taxation [55].

Moving on to the methodology, the SDDP method has been employed by numerous power companies for several years, as noted by Gjelsvik et al. [5]. Despite its widespread adoption, the method is not without drawbacks. As highlighted by several authors, it requires a long computation time. Furthermore, the power system in the thesis was scheduled primarily for revenue optimization. However, the model failed to incorporate maintenance time, which significantly impacts revenue. Within ProdRisk, there is a feature for both the module and the pump named 'maintenance'. This attribute makes it possible to implement capacity that is not available due to maintenance. Utilizing this attribute would have yielded a more realistic revenue outcome.

Further on, the primary limitation of the method is its lack of condition-dependent restrictions. This limitation became apparent when calculating pump efficiency using equation (3). The resulting values are displayed in figure A.4 in the appendix and show that the efficiency values are around 0.77 across all pump sizes. Initially, these values were set at 0.8 during the implementation of the pump. However, when calculating the efficiencies, the head H was determined based on reservoir levels and discharge by using extrapolation. The power values and discharge values were determined from the simulations. According to the physical formulas provided in the theory section 3.2, the efficiency should have been the same as initially given. Further on, SDDP simulations are executed twice; initially, the head coefficients are generated during the first run. These coefficients are then applied in the subsequent and final simulation. This process is defined as the initial run and the final run, respectively. This dual-run approach is necessary because condition-dependent restrictions mean that initially, there is no established head between the reservoir and the power plants. This is developed during the 50 main iterations, which consist of 50 forward simulations and 50 backward recursions. In the initial run, trajectories of the reservoir are established, from which the head is calculated. The condition-dependent restrictions in the program could explain the observed deviations in pump efficiency.

In summary, implementing a pump in the eastern part of RSK is projected to increase income from electricity sales compared to a system without a pump. A larger plant equipped with a bigger pump has a longer operation time, both for the turbine and pump. This is particularly notable during years of low inflow. While a larger pump may have lower efficiency, it enables the power plant to generate more electricity.

Another aspect worth exploring is running simulations for the largest pump under different price scenarios and conducting a sensitivity analysis for pricing. As

repeatedly mentioned in the thesis, economic factors drive investment decisions in new projects. The rise of renewable energy sources in the power market contributes to increased system unpredictability, leading to greater price fluctuations. Given this volatility, it would be beneficial to simulate different price scenarios using the model to enhance understanding and predict how the hydropower system might react. Such insights could improve producers' strategic positioning by equipping them with knowledge about potential future market conditions, thus better preparing them for future challenges.

It is conceivable to run the model across a spectrum of price scenarios, from high volatility to low, including intermediate scenarios, to examine how profitability varies. This analysis could inform decisions on whether investing in PHES systems, including larger or multiple installations, would be advantageous. In periods of high and short-term volatility, a larger pump capable of rapid response and maximum water pumping during low-price intervals might be necessary. Conversely, the model might reveal that new pump installation is constrained by limited reservoir capacity. For instance, while the reservoir at Sandvann is almost always full throughout the year, as shown in figure 8, Holmevann has considerable capacity. This suggests that installing a pump between Kvandalsfoss and Holmevann could be more viable. However, it would require constructing a new power station. Halting production at a hydropower station, especially for several years, suffers significant costs. Thus, implementing a pump at Kvanndal would require a production stoppage. On the other hand, continuing operations at Kvanndal while building a PHES between Kvanndalfoss and Holmevann might be the most profitable strategy for Lyse DA.

Building a new power station requires significant intervention in nature, involving tasks such as laying pipes and blasting mountains. In contrast, adding a pump to an existing power station is less invasive. However, considering the eastern system as a whole, the main limitation is the Kvanndalfoss reservoir, which has minimal capacity. To address the issue of an overfilled reservoir, one could either install a pump between Sandvann and Kvandalsfoss, as discussed in this thesis, or add a turbine at Suldal II to increase the discharge rate. Yet, installing a pump alone does not offer the same flexibility as a PHES system would. Numerous combinations and simulations need to be run to determine the most profitable solution. It is believable that Lyse DA has extensively used a scheduling optimization program for various simulations, concluding that constructing a new PHES between Kvanndalfoss and Holmevann is the most lucrative option. Moreover, it is plausible that their analyses considered the entire RSK system, including the western part, which opens up additional opportunities.

In Norway, most PHES systems operate seasonally, utilizing the pump during the summer months and the turbine during the winter. Additionally, several pumping

stations draw water from lateral reservoirs. By setting the equations (1) and (3) equal to each other and canceling out all components except for the heads and efficiencies, it becomes possible to determine the necessary height of the pump to balance the energy used for pumping with the energy generated by the turbine. This means the pumping head $H_{pumpmode}$ is set to be equivalent to the turbine mode head $Hturbinemode$, calculated as the height of the waterfalls multiplied by the product of the efficiencies of the turbine $\eta_{turbinemode}$ and the pump $\eta_{pumpmode}$. Pumping from lateral reservoirs consumes less energy than pumping from lower reservoirs for the plant. This makes it particularly cost-effective during periods of low power prices. Considering the simulation with the 40 MW pump, the turbine efficiency $\eta_{turbinemode}$ was 0.904 and the pump efficiency was $\eta_{pumpmode}$ is 0.773. If the turbine head $H_{turbinemode}$ is set at the nominal head of 314.8 m, the required pumping head is 220.0 m. This indicates that the pumping height needs to be 70% of the turbine mode head for the energy used in pumping to equal the energy generated by the turbine. It is also worth considering a scenario where the lateral reservoir pumping head is only 50% of the turbine mode head, which would mean even less energy is required for pumping compared to what the turbine generates. This could be significantly more profitable than pumping from the lower reservoir. The calculations is found in section A.2 in the Appendix.

Lastly, it would be interesting to explore how the eastern part of RSK with PHES would perform in selling power on both the spot and reserve markets. With the flexibility and storage capacity that PHES provides, it potentially offers significant opportunities for selling power to the reserve market. According to the study by Helseth et al. *Optimal Medium-Term Hydropower Scheduling Considering Energy and Reserve Capacity Markets* [23], introducing a reserve capacity market alongside the spot market proved to be profitable.

Furthermore, Statnett [56] indicates that the demand for fast response reserves FFR is particularly high during periods of low load and production, especially when there is a substantial proportion of wind power and imports. In such situations, PHES could serve as an effective tool by utilizing the pump for consumption and harnessing energy from wind farms. Statnett emphasizes that these conditions are most commonly observed during the summer, particularly in years with low hydrological levels. This highlights even more the importance and necessity of PHES. In years when hydrological levels are low, a pump can enable the reuse of water, enhancing resource efficiency.

This further extends to considerations of climate change. According to the Meteorological Institute, when comparing the weather period from 1991-2020 to the earlier period of 1961-1990, there is a 7% increase in annual precipitation. Additionally, rainfall is heavier on wet days. Given the limited reservoir capacity of Kvandalsfoss,

it is likely that the turbines at Suldal II will not be able to manage all the water inflow. However, if a pump were installed to transfer water to either Holmevann or Sandvann, much more of the water could be utilized, minimizing losses to bypass flows [57].

As highlighted in the introduction, PHES is a crucial regulatory renewable energy source for the future. It allows for the later use of wind and solar energy, ensuring that their energy is not wasted. As discussed earlier, PHES can transform energy of low societal benefit into energy of higher societal value. This is particularly valuable during years of low inflow, as the flexibility provided by PHES can significantly benefit society.

Lastly, unlike wind and solar power, PHES is a renewable energy source that can be regulated. Furthermore, since PHES has been developed alongside conventional hydropower systems, converting existing hydropower stations to PHES has a smaller impact on nature compared to the development of new wind and solar power farms.

7 Conclusion

The thesis explored the operational impacts of integrating PHES into the Røldal-Suldal hydropower system and assessed how different pump sizes affected system performance. It also evaluated the economic benefits of PHES integration, focusing on identifying the pump size that maximized revenue and offered the greatest economic advantage.

In the eastern part of RSK, larger pump sizes result in higher revenue. With a bigger pump, the operational time for both the turbine and the pump increases, thereby boosting production. A larger pump moves and utilizes more water, which accounts for the extended operational times. However, with higher discharge, the turbine efficiency decreases. Specifically, the turbine efficiency with the 40 MW pump was lower compared to turbines equipped with smaller pumps. To determine the profitability of investing in a pump in the eastern part of RSK, further calculations of investment costs are necessary. However, the simulation with the biggest pump, 40 MW achieved the highest revenue.

Incorporating PHES into the energy system allows for the storage of water in reservoirs, which can be utilized during periods of high consumption and low output from solar and wind power sources. As such, PHES can transform energy of low societal benefit into energy with greater societal value. A key advantage of PHES is its flexibility and the supply security it offers. As today's power market transitions from fossil fuels to renewable energy sources, this shift presents challenges, particularly because solar and wind are intermittent energy sources. As the European power market transitions from fossil fuels to renewable energy sources, it faces challenges due to the intermittent nature of solar and wind energy. Norway, with its vast reservoir capacity, serves as Europe's green battery. Norwegian shares a common power market with the other Nordic countries. The Nordic market is increasingly integrating with the broader European power market. Therefore, it is crucial to have PHES systems in place to store surplus energy for later use. PHES offers multiple benefits to the power grid, including energy storage, enhanced grid stability, seamless integration of renewable energies, and effective load balancing.

PHES systems offer large-scale energy storage abilities. PHES systems respond quickly to fluctuations in supply and demand, thus effectively maintaining grid stability. PHES systems effectively integrate intermittent renewable energy sources like wind and solar by storing and releasing energy as needed, thus facilitating their broader integration into the grid. PHES systems are good tools for load balancing, enhancing the power grid's overall efficiency by shifting electricity generation from periods of low demand to times of high demand.

With the numerous advantages that PHES offers, it is clear that this technology is a key driver in addressing the climate crisis. By enabling a shift towards entirely renewable energy sources, PHES can play a crucial role in achieving the goal of limiting global warming to 1.5^{o}C.

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A Appendix

A.1 Figures

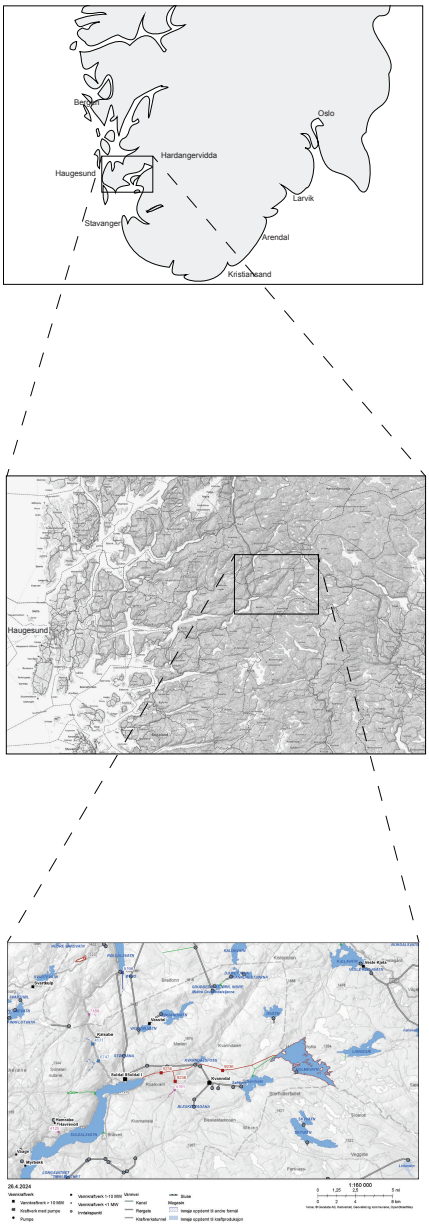


Figure A.1: Geographical location of the Røldal-Suldal Kraft power stations (RSK) in Norway. The figure has been adapted and modified from [17].

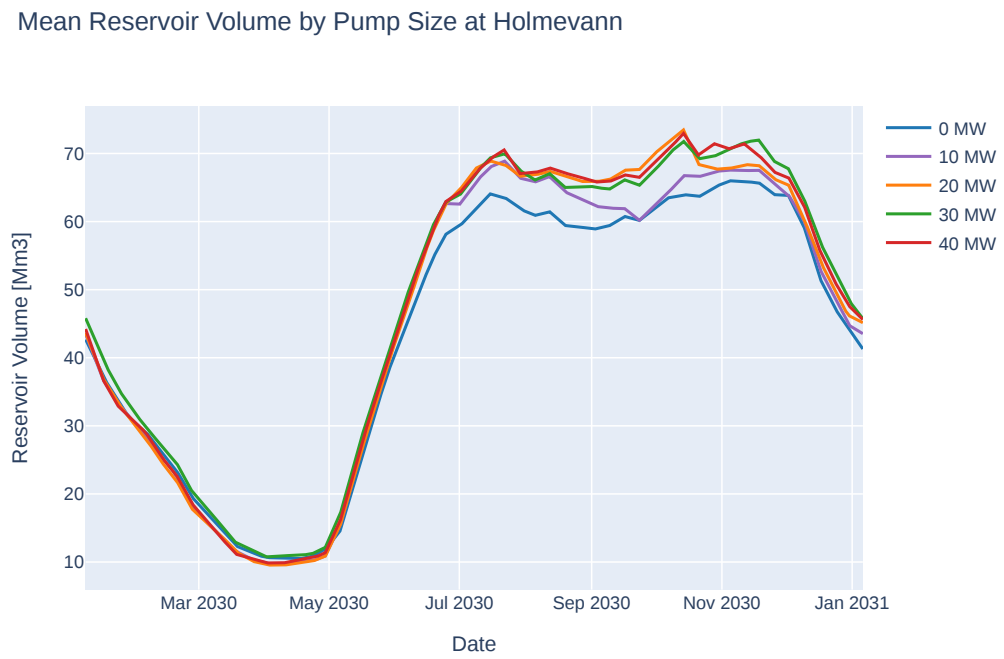


Figure A.2: Reservoir level for Holmevann for various pump sizes. The y-axis indicates the reservoir level in a million cubic meters, while the x-axis tracks the dates throughout the years 2030 to 2031. Each line on the graph represents a different pump capacity, including a baseline scenario labeled as '0 MW', which shows the reservoir levels without any pump intervention

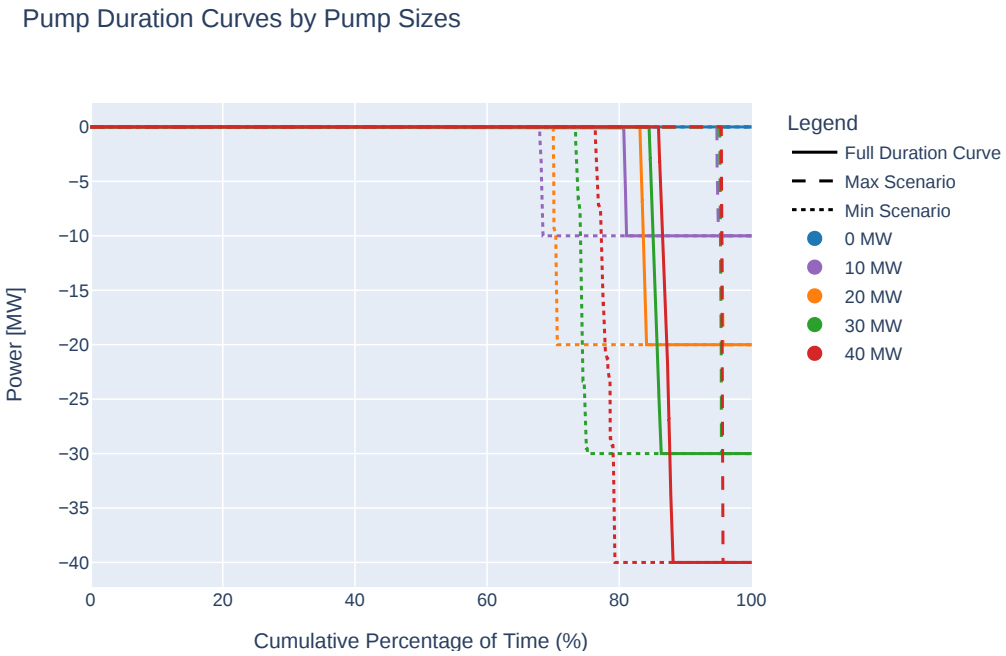


Figure A.3: Duration curves for the pump at plant Kvanndal for various pump sizes. The three duration curves represent data from a year with minimum power output, an aggregate 30-year period, and a year with maximum power output. The y-axis indicates the power output, given in millions of Watts, while the x-axis gives the total percentage of time for a year in percentage.

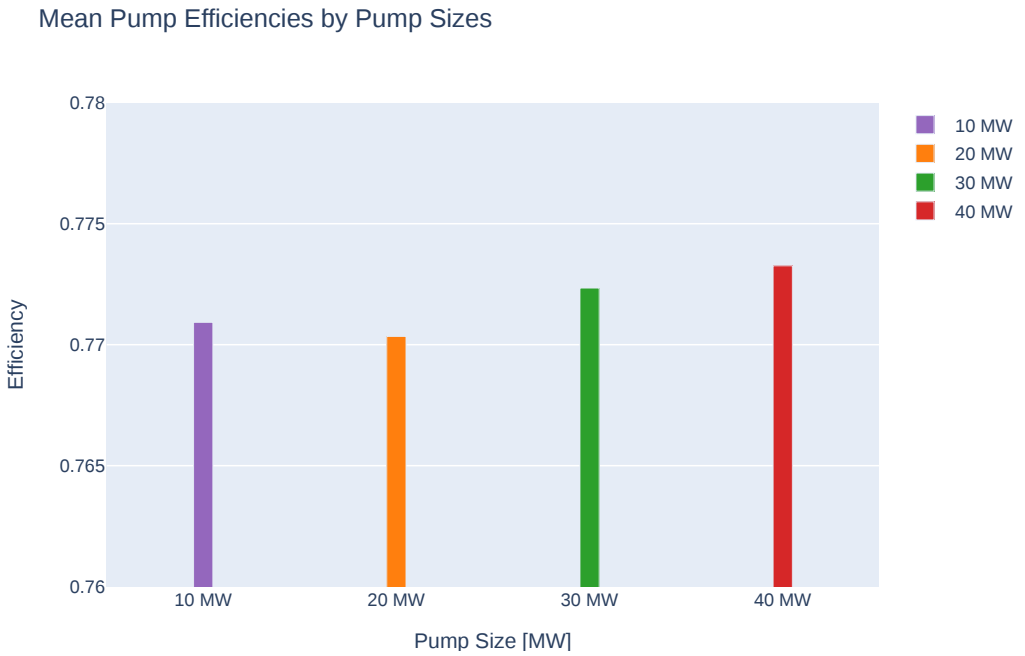


Figure A.4: Mean pump efficiencies for various pump sizes. The efficiency was set at 0.8 for all pump sizes. However, when recalculated using equation (3), the efficiency values exhibited a slight decrease. The y-axis indicates the efficiency, while the x-axis gives the pump sizes.

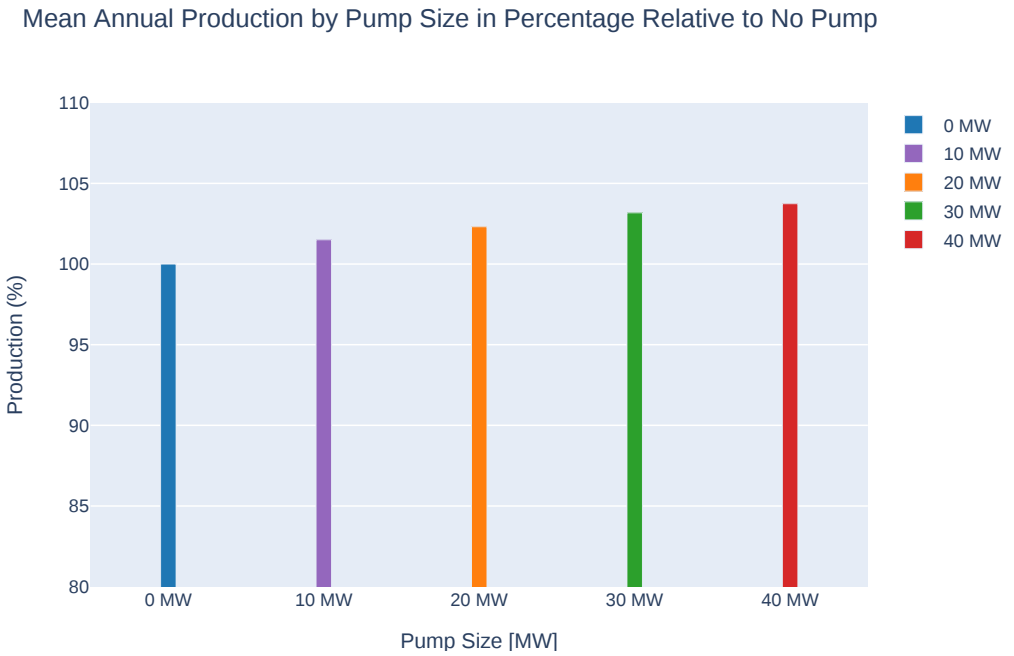


Figure A.5: Mean annual production in percentage relative to no pump for various pump sizes. The y-axis displays the production from the power stations Kvanndal and Suldal II, given in percentage. Here, the production from the baseline simulation, without a pump, is set to 100%. The production for simulations with different pump sizes is presented as a percentage compared to this baseline. The x-axis gives the pump sizes.

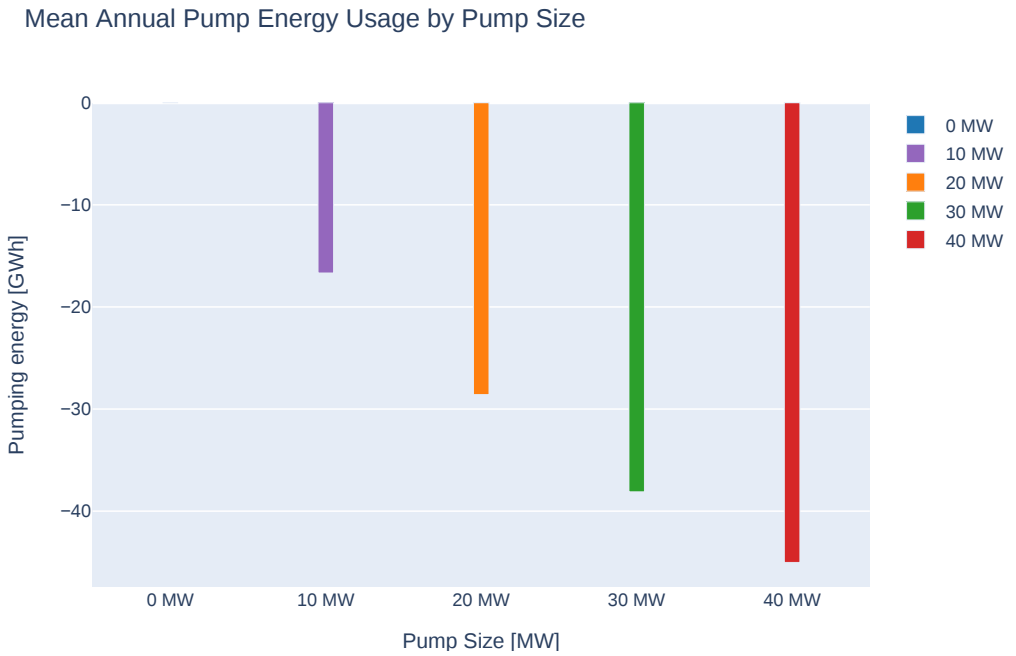


Figure A.6: Mean annual energy used for pumping for various pump sizes. The y-axis displays the mean annual energy required for pumping operations, given in gigawatt-hours.

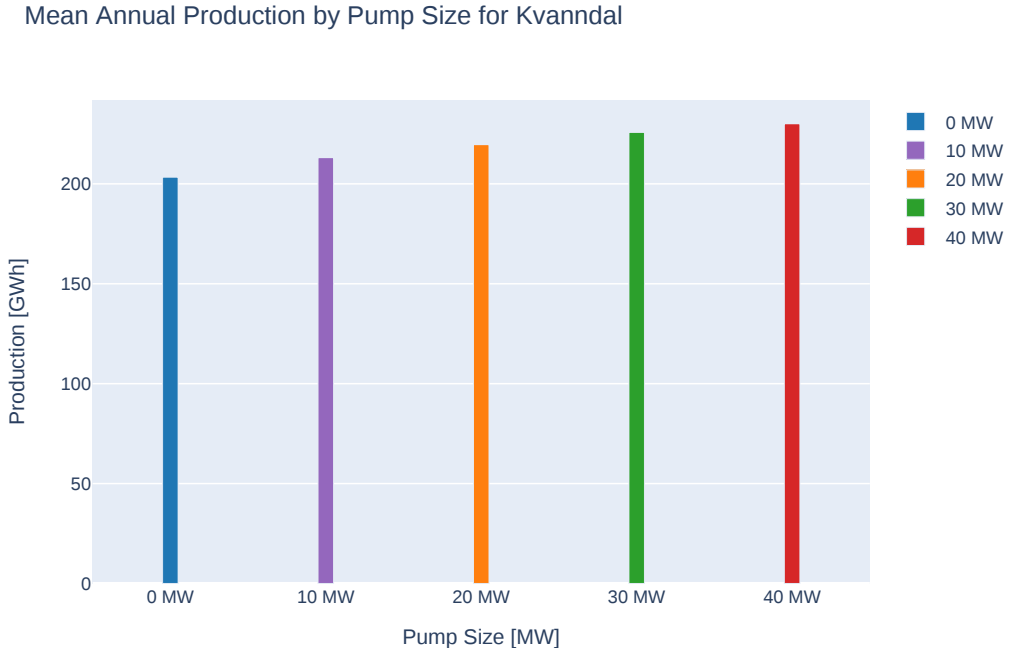


Figure A.7: Mean annual production at Kvanndal power station for various pump sizes. The y-axis indicates the mean annual energy production in gigawatt-hours, while the x-axis gives the pump sizes.

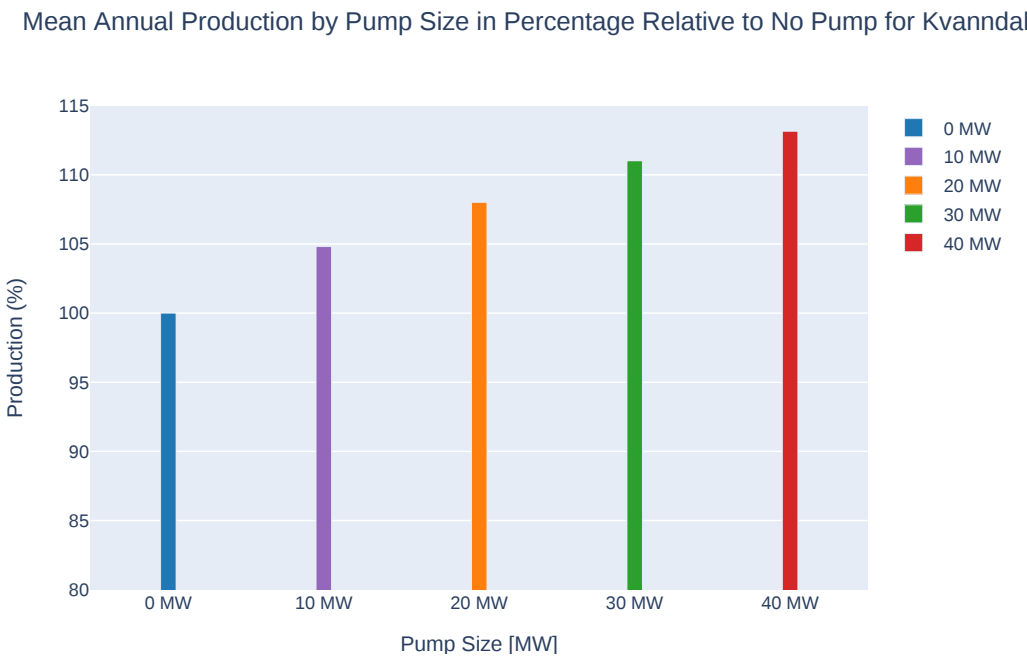


Figure A.8: Mean annual production at Kvanndal power station in percentage relative to no pump for various pump sizes. The y-axis displays the production from the power station Kvanndal, given in percentage. Here, the production from the baseline simulation, without a pump, is set to 100%. The production for simulations with different pump sizes is presented as a percentage compared to this baseline. The x-axis gives the pump sizes.

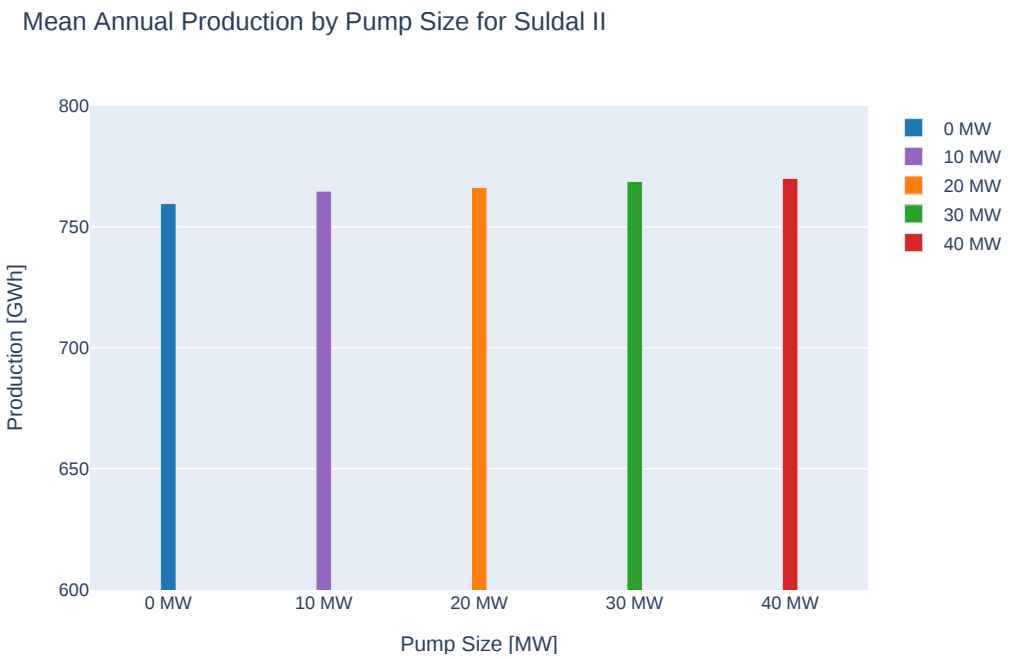


Figure A.9: Mean annual production at Suldal II power station for various pump sizes. The y-axis indicates the mean annual energy production in gigawatt-hours, while the x-axis gives the pump sizes.

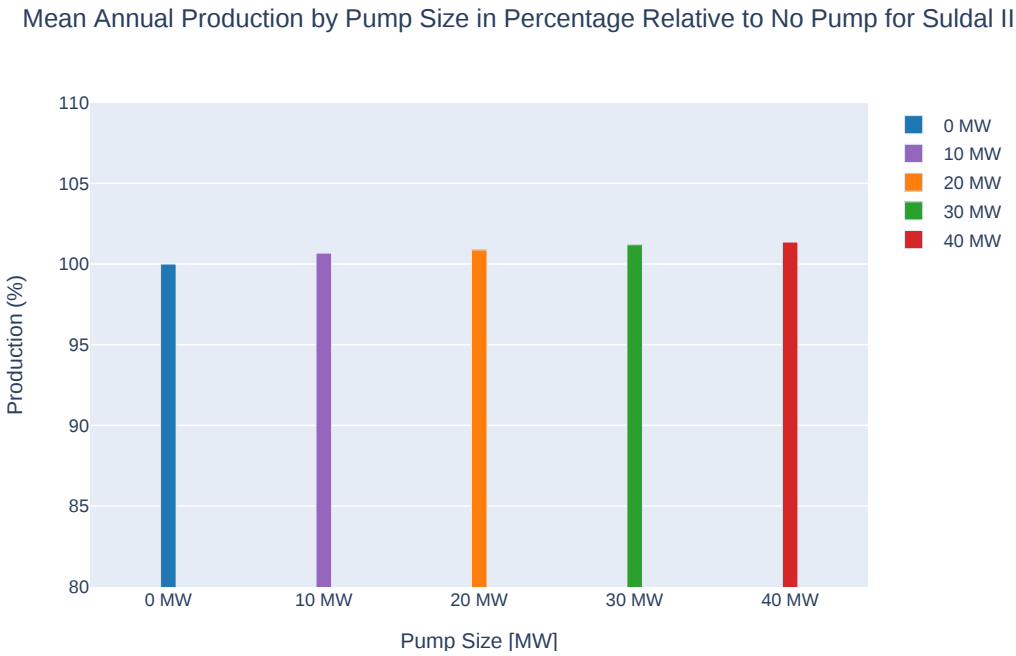


Figure A.10: Mean annual production at Suldal II power station in percentage relative to no pump for various pump size. The y-axis displays the production from the power station Suldal II, given in percentage. Here, the production from the baseline simulation, without a pump, is set to 100%. The production for simulations with different pump sizes is presented as a percentage compared to this baseline. The x-axis gives the pump sizes.

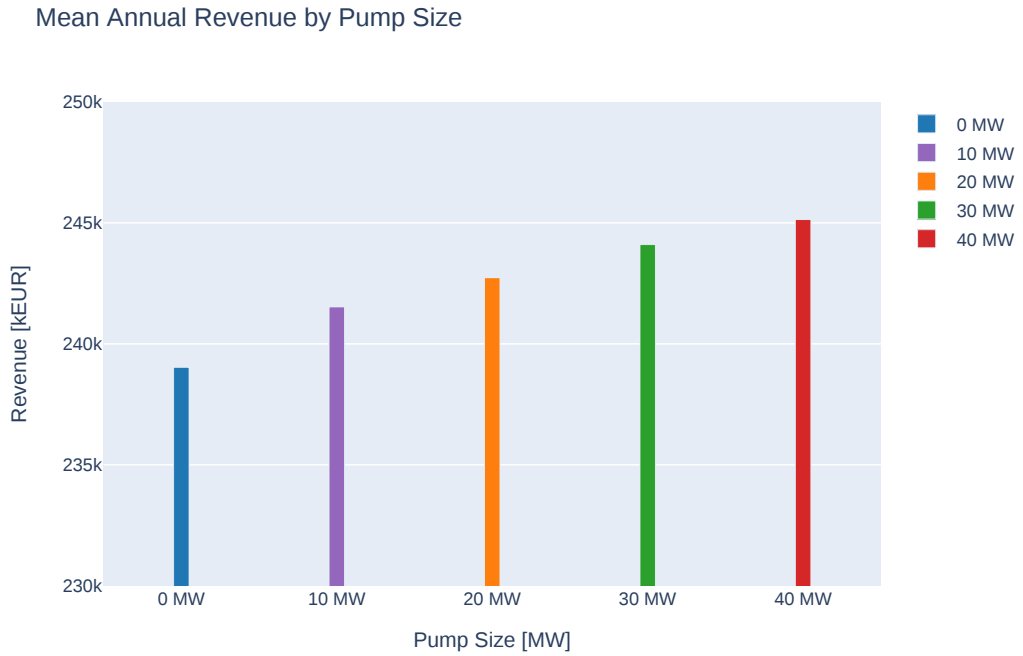


Figure A.11: Mean annual revenue for Kvanndal and Suldal power stations for various pump sizes. The y-axis displays the revenue in millions of euros, while the x-axis gives various pump sizes.

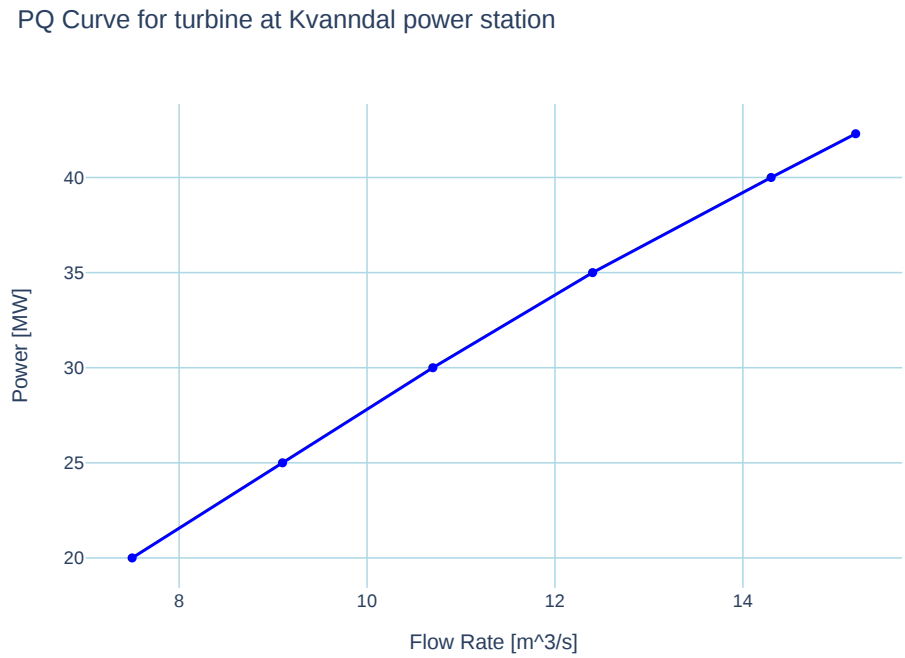


Figure A.12: The PQ curve for the turbine at Kvanndal power station illustrates the relationship between power output and discharge. Power output, measured in megawatts is displayed on the y-axis. The corresponding discharge, in cubic metres per second, is shown on the x-axis. The curve is relatively flat, though it exhibits a slightly steeper slope at lower values of power and discharge.

A.2 Calculations

Calculations of the required head from lateral reservoirs to equate the energy used for pumping with the energy generated by the turbine. To calculate the required head, equation (1) and equation (3) are equated which is shown in equation (10):

$$P_{turbine} = P_{pump}. \tag{10}$$

The parameters for calculating the power of both the turbine and the pump are incorporated into equation (11):

$$Q \cdot \rho_{water} \cdot g \cdot H_{turbinemode} \cdot \eta_{turbinemode} = \frac{Q \cdot \rho_{water} \cdot g \cdot H_{pumpmode}}{\eta_{pumpmode}}. \tag{11}$$

The components Q, ρ_{water} and g are eliminated, simplifying the expression to equation (12):

$$H_{turbinemode} \cdot \eta_{turbinemode} = \frac{H_{pumpmode}}{\eta_{pumpmode}}. \tag{12}$$

Resolving equation (12) for $H_{pumpmode}$ and substituting the turbine efficiency, $\eta_{turbinemode}$ to 0.904 and the pump efficiency, $\eta_{pumpmode}$ to 0.773. Also, setting the

turbine mode head $H_{turbinemode}$ to the nominal head of 314.8 m. This leads to equation (14):

$$H_{pumpmode} = 314.8\text{m} \cdot 0.904 \cdot 0.773. \tag{13}$$

By multiplying the efficiencies together, one obtains:

$$H_{pumpmode} = 314.8\text{m} \cdot 0.698. \tag{14}$$

Solving equation (14) yields a required pumping head of 220.0 m. This result implies that the pumping head needs to be 70% of the turbine mode head to balance the energy expended in pumping with the energy generated by the turbine.



Norges miljø- og biovitenskapelige universitet
Noregs miljø- og biovitskaplege universitet
Norwegian University of Life Sciences

Postboks 5003
NO-1432 Ås
Norway